

Triples of Consecutive Powerful Numbers: Pell Addresses, a Height Inequality, and Witnesses

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October 4, 2026

Abstract

We study the question of Erdős whether three consecutive integers can all be powerful. The middle n of such a triple is a Pell number $T_k(m)$ at a lattice point (m, k) with a squarefree kernel $m \equiv 7 \pmod{8}$. We prove a height inequality, $\log_{10} T_k(m) \geq m'(1.5 \log_{10} m - \log_{10} m') - \log_{10} 2$, which splits the kernels into two wings by whether m divides $U_1(m)$, and we recast the reduction of Mollin and Walsh as a statement about witnesses, primes that divide a Pell number exactly once, organized by rank. We compute: all 445 lattice points with kernel up to 10^8 and $T_k < 10^{2000}$ are excluded by parity or by a certified witness, and with the height inequality and Reinhart's list of the kernels dividing U_1 up to $1.5 \cdot 10^{12}$ this shows that no triple has a middle below $1.83 \cdot 10^{18}$; further regions are excluded whatever the size of the middle; and of the 961 building blocks M_d of these lattice points, 954 are shown not to be powerful by a certified witness, a prime proved prime that divides the block exactly once, and 7 remain open. A profile theorem collects the conditions on a counterexample that we prove or compute, and the weak point of the bound, the kernels dividing U_1 , is stated as an open problem. Every computed number is generated from the deposited data or from a cited constant, and every numbered statement is labeled as proved, taken from the literature, computed, measured, heuristic, or open.

About this series. This paper is one of three parts that share one source, one bibliography, and one numbering. A statement numbered I.4.5 carries that number in its part and in the collected volume. Every number that we computed is generated from the result files of the data deposit or from a cited constant; numbers from the literature carry a citation. The ranges and parameters of a computation and the constants of an argument are written in the text. The open questions of all parts are collected as Problems A to I in the framework introduction of the collected volume and in the file `OPEN_PROBLEMS.md` of the data deposit; both are generated from the parts. This part is a preprint. The author corrects any error that becomes visible after the release in a new version of the record; earlier versions remain available under their own version numbers.

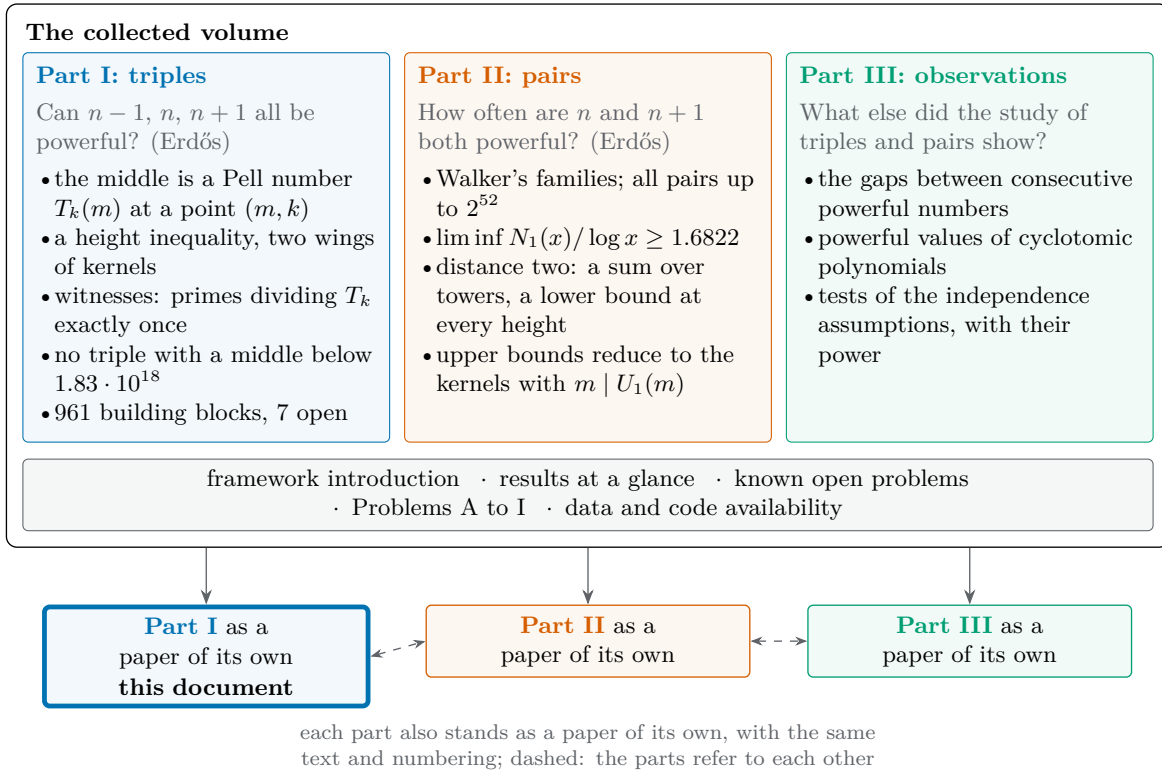


Figure 1: **What this research did.** It followed two questions of Erdős on consecutive powerful numbers, on triples and on pairs, through Pell equations, proved or computed what it could reach, and recorded the rest as observations and open problems. The collected volume contains the three parts and the framework that ties them together; each part also stands as a paper of its own, with the same text and numbering, and the parts refer to each other. The heavier frame marks the document in hand.

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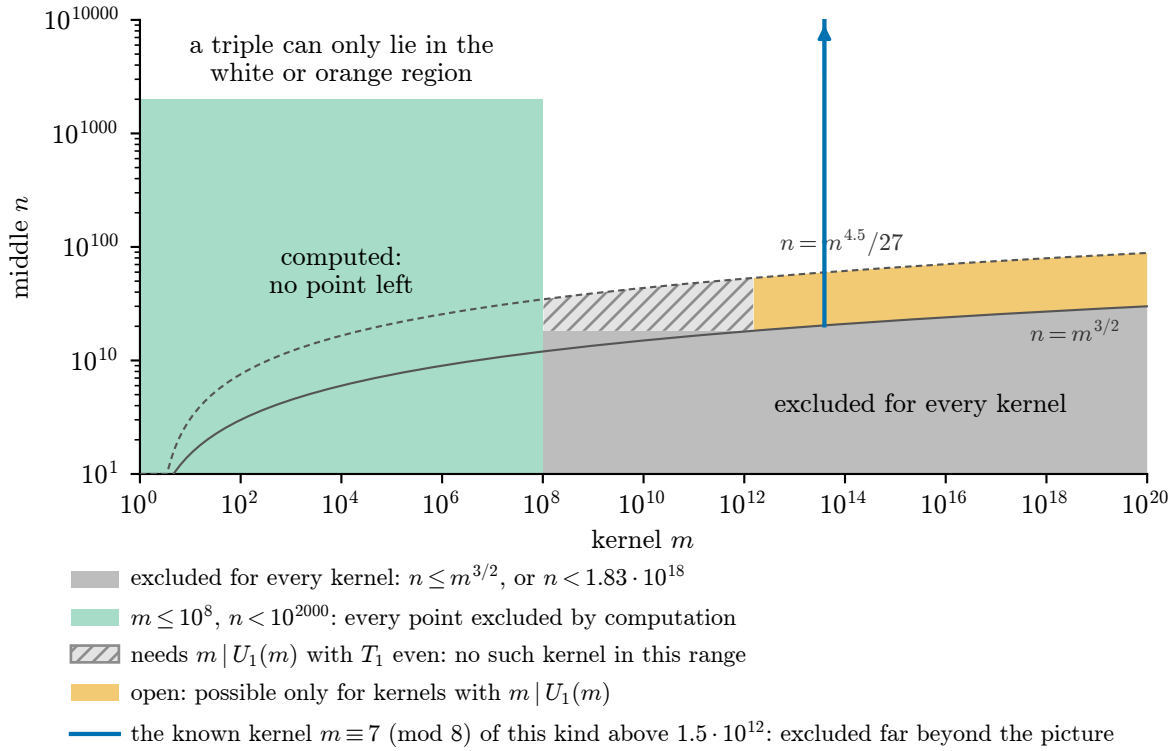


Figure 2: **Where a triple could still lie.** The kernel m on a logarithmic scale, the middle n on a doubly logarithmic one. Gray: excluded for every kernel, by the height bound of Corollary I.2.3 or by Proposition I.6.2(d). Hatched: here a middle needs $m \mid U_1(m)$ with $T_1(m)$ even, and Reinhart’s complete list has no such kernel between 10^8 and $1.5 \cdot 10^{12}$ (Remark I.2.5; the parity of T_1 is computed in Proposition I.6.3). Green: the box of Theorem I.3.1, where every lattice point is excluded by computation. Orange: the weak point of the bound, kernels with $m \mid U_1(m)$ above $1.5 \cdot 10^{12}$ (Question I.9.1); the one known kernel of this kind above $1.5 \cdot 10^{12}$ in class 7 with T_1 even, drawn as a line, admits no middle below the height given in Table 3, and so does the other known one, which lies in the box. The picture does not draw the classes of kernels that Section 5 excludes, so a triple could lie only in parts of the white and orange regions.

1 Introduction and the address system

A positive integer is *powerful* if every prime that divides it divides it at least twice. Erdős asked whether three consecutive integers can all be powerful, and conjectured that they cannot [23, 10]; on the attribution among Erdős’s papers see [3, Section 7]. Mollin and Walsh conjectured it in a form about Pell numbers [45, p. 111] (Section 4). Two consecutive powerful numbers exist in abundance, since $x^2 - 8y^2 = 1$ has infinitely many solutions, and four consecutive ones are impossible, since one of four consecutive integers is 2 modulo 4. No triple is known. The *abc* conjecture implies that there are only finitely many, and its explicit

forms give finite bounds (Remark I.4.12); unconditionally the question is open. Bloom records that no triple has a middle below $7.38 \cdot 10^{28}$ by OEIS A076445 [10, 47], a statement that rests on the listed terms of that sequence (Section 6).

We expect that there is no triple, and we give the reasons as evidence, not as proof. First, the *abc* conjecture implies that there are at most finitely many (Remark I.4.12). Second, none has been found: none has a middle below $1.83 \cdot 10^{18}$ (Proposition I.6.2(d)), and none lies below 10^{22} by an enumeration of pairs that is described as exhaustive but is not certified [48, 43]. Mian and Siddique prove, with machine-checked theorems in Lean 4, that no triple lies below 10^{14} [43]; Table 1 lists these bounds with what each rests on. Third, a triple needs a powerful Pell number $T_k(m)$, and this means that for each divisor $d \geq 5$ of k with $k < d(4d - 1)$ every prime of rank d , and there is at least one, must divide $T_d(m)$ at least twice (Corollary I.4.8); at our 445 lattice points the median number of such conditions is 3 (Proposition I.6.6). In the data, a prime p divides the first Pell term that it divides at least twice about as often as the model with probability $1/p$ predicts (Proposition I.6.1); under that model each condition is one more rare event. The third reason is a heuristic, and the model is not proved.

no triple below	what the bound rests on	source
10^{14}	machine-checked theorems in Lean 4	[43]
$1.83 \cdot 10^{18}$	computation in this part, every witness proved prime, together with Reinhart’s computer proof that his list of the kernels dividing U_1 is complete up to $1.5 \cdot 10^{12}$	Proposition I.6.2(d)
$3.88 \cdot 10^{20}$	the same, if Reinhart’s extended search missed nothing	Proposition I.6.2(d)
10^{22}	an enumeration of the pairs, described as exhaustive, not certified	[48, 43]
$7.38 \cdot 10^{28}$	the listed terms of OEIS A076445; the entry does not state that the list is complete	[10, 47]

Table 1: **Bounds below which no triple of consecutive powerful numbers is known, and what each rests on.** Our two rows and the last one bound the middle of a triple; the rows for 10^{14} and 10^{22} are stated for the triple, whose members differ from its middle by at most one.

The problem has an exact address system, due to Mollin and Walsh, Walker and Sentance (Lemmas I.1.1 to I.1.4): the middle n of a triple is a term $T_k(m)$ of the Pell sequence of a squarefree kernel $m \equiv 7 \pmod{8}$, at an odd index k that is a multiple of the part m' of m prime to U_1 . So a triple is a lattice point (m, k) at which $T_k(m)$ is even and powerful, and the question becomes a question about the prime factors of Pell numbers. Two facts make it hard. Granville showed that the nonexistence of triples implies that there are infinitely many primes p with $p^2 \nmid 2^p - 2$ (Theorem I.4.9), a statement that is not known unconditionally; so a proof would also settle an open problem about Wieferich primes. And an *abc* inequality with exponent below $4/3$ would exclude all large triples, while the exponents derived from the explicit form of the conjecture, itself unproved, are 1.7 and more (Remark I.4.12).

This part does three things. It proves a height inequality that bounds the middle of a triple from below in terms of its kernel, and splits the kernels into two wings by whether m divides $U_1(m)$. It recasts the reduction of Mollin and Walsh as a statement about witnesses,

primes that divide a Pell number exactly once, and organizes the witnesses by their rank. And it computes: every lattice point with kernel up to 10^8 and $T_k < 10^{2000}$ is excluded by a certified witness or by parity, and of the 961 building blocks M_d that occur there, 954 are shown not to be powerful by a certified witness, a prime that divides the block exactly once and is proved prime; 7 remain open. The weak point of the bound is wing (ii), the kernels with $m \mid U_1$: for them only $n > m^{3/2}$ is available, all such kernels up to $1.5 \cdot 10^{12}$ are known [52, Remark 5.5], up to $5.325 \cdot 10^{13}$ with a caveat of the author [53], and beyond that we know no lower bound for their middles other than $n > m^{3/2}$ (Section 2). Part II meets the kernels with $m \mid U_1(m)$ and $T_1(m)$ even, of both residue classes, from the side of pairs, and the collected volume lists the open problems of both parts together.

Main results.

- *The height inequality* (Theorem I.2.1): at every lattice point,

$$\log_{10} T_k(m) \geq m'(1.5 \log_{10} m - \log_{10} m') - \log_{10} 2.$$

With Remark I.2.2, a triple with middle below 10^H has its kernel in one of two wings (Corollary I.2.4): $m \nmid U_1$ and $m < (27 \cdot 10^H)^{2/9}$, or $m \mid U_1$ and $m < 10^{2H/3}$.

- *The reduction theorem* (Theorem I.3.1): among the 10 132 126 kernels up to 10^8 , exactly 445 lattice points have $T_k < 10^{2000}$; 74 of them have T_k odd, and each of the other 371 has a certified witness below $2 \cdot 10^5$. So no triple has a middle $n < 10^{2000}$ with $\text{sqfree}(n^2 - 1) \leq 10^8$, and no triple at all has a middle below $1.83 \cdot 10^{18}$ (Proposition I.6.2(d)).
- *The witness statement* (Theorem I.4.6): there is no triple if and only if every lattice point with $T_1(m)$ even has a witness, and a witness can then be found among the primes whose rank divides k with odd cofactor; a powerful T_k would also have to satisfy the exponent-3 condition of Proposition I.4.4. The *b*-box (Theorem I.4.5) excludes every middle, of any size, with kernel up to 10^6 and squarefree part up to 10^4 ; the strip (Theorem I.5.1) shows that a triple with kernel up to 2000 has a middle of more than $8.4 \cdot 10^7$ digits; the smooth kernels (Theorem I.5.3) exclude a further range below 10^{2000} .
- *The building blocks* (Proposition I.6.8): of the 961 primitive parts M_d on the 79 kernels of the lattice points, 954 have a certified witness; 7 are open, each with at least 173 digits. Theorem I.7.1 collects the conditions on a counterexample that this part proves or computes.

How to read this part. Every statement is followed by a paragraph *In words* that says what it means with as few formulas as possible; a reader who wants the results can read the statements and these paragraphs. A reader who wants to check can read the proofs, the paragraphs *The computation*, and the tables; every number we computed is generated from the data files of the deposit or from a cited constant, and every numbered statement carries a label that says whether it is proved, taken from the literature, computed, measured, a heuristic, or open (see the table at the end). A statement that combines several kinds carries the label of the part that needs the most trust: a proof that uses our own computation is labeled computed, and a proof that uses a published result is labeled proved. Table 2 collects the notation.

Plan. The rest of this section sets up the notation and the address system. Section 2 proves the height inequality and the two wings. Section 3 states the reduction theorem and describes the computation behind it. Section 4 proves the exponent condition and the reduction to a witness statement, computes the b -box, and records Granville's theorem. Section 5 adds the strip, the smooth kernels and the square-type middles. Section 6 reports the data: the exponents at the rank, the list of pairs at distance two, the ladders for the known kernels dividing U_1 , the witnesses, and the building blocks. Section 7 collects the profile of a counterexample, Section 8 records the routes that do not close the problem, and Section 9 states the open questions.

1.1 Notation

A positive integer n is *powerful* if p^2 divides n for every prime p that divides n . For a squarefree integer $m \geq 2$ let $\varepsilon_m = T_1 + U_1\sqrt{m}$ be the fundamental solution of $x^2 - my^2 = 1$, that is, the solution in positive integers with the smallest x . For $k \geq 1$ write

$$\varepsilon_m^k = T_k + U_k\sqrt{m},$$

and $T_k(m)$, $U_k(m)$ when the kernel m has to be named. Put

$$m' = \prod_{p|m, p \nmid U_1} p,$$

the product of the primes of m that do not divide U_1 . For an integer $N \geq 1$ let $\text{sqfree}(N)$ be the product of the primes that divide N to an odd power.

In words. A number is powerful when every prime in it appears at least twice; for example $72 = 2^3 \cdot 3^2$ is powerful and $12 = 2^2 \cdot 3$ is not. The equation $x^2 - my^2 = 1$ is Pell's equation. Its smallest solution ε_m produces all other solutions as powers, and T_k and U_k are the two parts of the k -th power. The number m' collects the primes of m that do not divide U_1 .

1.2 The address system

Lemma I.1.1 (Mollin and Walsh [45, Theorem, p. 110]). [literature] *There are three consecutive powerful numbers if and only if there are a squarefree $m \equiv 7 \pmod{8}$ and an odd k such that $T_k(m)$ is an even powerful number and m divides $U_k(m)$. In that case the triple is $(T_k - 1, T_k, T_k + 1)$, and $m = \text{sqfree}(T_k^2 - 1)$.*

Mollin and Walsh state the criterion with the fundamental unit of $\mathbb{Q}(\sqrt{m})$. For $m \equiv 7 \pmod{8}$ this unit is ε_m : the ring of integers is $\mathbb{Z}[\sqrt{m}]$, and m has a prime factor $\equiv 3 \pmod{4}$, so no unit of norm -1 exists.

In words. The middle n of a triple of consecutive powerful numbers is always even, and it is a Pell number T_k for exactly one kernel m . So the search for triples becomes a search over pairs (m, k) .

Lemma I.1.2 (Walker [71, Theorems 2.2(3) and 2.5]; Mollin and Walsh [45, p. 111]). [literature] *Let m be squarefree and let p be a prime dividing m . Then $p \mid U_k$ if and only if $p \mid kU_1$. Hence $m \mid U_k$ if and only if $m' \mid k$.*

Proof. By the binomial theorem, $U_k = \sum_{i \text{ odd}} \binom{k}{i} T_1^{k-i} U_1^i m^{(i-1)/2}$. For $p \mid m$ every term with $i \geq 3$ is divisible by p , so $U_k \equiv k T_1^{k-1} U_1 \pmod{p}$. From $T_1^2 - m U_1^2 = 1$ we get $T_1^2 \equiv 1 \pmod{p}$, so T_1 is invertible modulo p , and the first claim follows. A prime $p \mid m$ with $p \mid U_1$ divides every U_k ; a prime $p \mid m$ with $p \nmid U_1$ divides U_k exactly when $p \mid k$. Since m is squarefree, this gives the second claim. $\square$

In words. Whether m divides U_k depends only on whether k is a multiple of m' . So for each kernel the useful indices k are exactly the multiples of m' .

Lemma I.1.3 (Sentance [60, Theorem and Converse, p. 272]). [literature] *Let $n \equiv 0 \pmod{4}$. Then $n - 1$ and $n + 1$ are both powerful if and only if $n = T_k(m)$ for a squarefree $m \equiv 7 \pmod{8}$ and an odd k with $m' \mid k$.*

Proof. ($\Leftarrow$) We have $n^2 - 1 = m U_k^2$ and $m \mid U_k$ by Lemma I.1.2, so $n^2 - 1 = m^3 w^2$ for an integer w . As n is even, $n - 1$ and $n + 1$ are coprime, so for every prime p the full power of p in $m^3 w^2$ lies in one of them. Hence each of $n \pm 1$ has the form $a^3 s^2$ with $a \mid m$ squarefree, and such a number is powerful. ($\Rightarrow$) If $n \pm 1$ are powerful and coprime, then $N = n^2 - 1$ is powerful. Write $N = a^2 b^3$ with b squarefree; then $N = b(ab)^2$ with $b \mid ab$. So (n, ab) solves $x^2 - by^2 = 1$, and $(n, ab) = (T_k, U_k)$ for $m = b$ and some $k \geq 1$. From $n \equiv 0 \pmod{4}$ we get $n^2 - 1 \equiv 7 \pmod{8}$; as U_k is odd, $m \equiv 7 \pmod{8}$. An even T_k forces k odd, since $T_{2j} = 2T_j^2 - 1$ is odd. Finally $m \mid U_k$ gives $m' \mid k$ by Lemma I.1.2. $\square$

Lemma I.1.4 (Sentance [60, p. 272]; Mollin and Walsh [44, Lemma 1.1]). [literature] *(General even middle.) Let $n \geq 2$ be even. Then $n - 1$ and $n + 1$ are both powerful if and only if $n = T_k(m)$ for a squarefree odd $m \geq 3$ and an odd k with $m' \mid k$ and $T_1(m)$ even. The pair (m, k) is unique: $m = \text{sqfree}(n^2 - 1)$, and k is determined because T_k increases with k . Moreover $m \equiv 7 \pmod{8}$ if and only if $n \equiv 0 \pmod{4}$, and $m \equiv 3 \pmod{8}$ if and only if $n \equiv 2 \pmod{4}$; kernels $m \equiv 1, 5 \pmod{8}$ never occur. An odd n never occurs.*

Proof. As for Lemma I.1.3; only the residues change. For $n \equiv 2 \pmod{4}$ we get $n^2 - 1 \equiv 3 \pmod{8}$, and with U_k odd, $m \equiv 3 \pmod{8}$. For odd k we have $T_k \equiv T_1 \pmod{2}$: if T_1 is even, every term of $T_k = \sum_j \binom{k}{2j} T_1^{k-2j} (m U_1^2)^j$ contains a factor T_1 ; if T_1 is odd, every term with $j \geq 1$ contains the even factor $m U_1^2 = T_1^2 - 1$. So T_k is even exactly when T_1 is. If n were odd, $n \pm 1$ would be even powerful numbers, hence both divisible by 4, which is impossible at distance two. $\square$

The residue split and the uniqueness statement are added here; the rest is Sentance's theorem.

We call a pair (m, k) with $m \equiv 7 \pmod{8}$ squarefree and k an odd multiple of m' a *lattice point*. By Lemma I.1.3, the lattice points with T_k even are exactly the pairs of powerful numbers at distance two whose middle is $\equiv 0 \pmod{4}$. By Lemma I.1.1, a triple needs one more condition: T_k itself must be powerful.

In words. Two powerful numbers that differ by 2 always sit around a Pell number T_k . So every such pair has an address (m, k) , and each address carries at most one pair. A triple is a pair whose middle is powerful as well. The kernel also tells the remainder: $m \equiv 7 \pmod{8}$ exactly when the middle is a multiple of 4, and $m \equiv 3 \pmod{8}$ when it is 2 more than a multiple of 4.

1.3 Notation and terms at a glance

Table 2 collects the symbols and words used in this part, with the place where each is defined.

symbol or term	meaning	defined in
powerful	every prime factor occurs at least twice	§1
$v_p(N)$	the exponent of the prime p in N	standard
$\text{sqfree}(N)$	the product of the primes that divide N to an odd power	§1
kernel m	a squarefree $m \geq 2$; for a middle n it is $m = \text{sqfree}(n^2 - 1)$	Lemma I.1.4
ε_m, T_k, U_k	the smallest solution $T_1 + U_1\sqrt{m}$ of $x^2 - my^2 = 1$, and its k -th power $T_k + U_k\sqrt{m}$	§1
m'	the product of the primes of m that do not divide U_1	§1
address (m, k)	the pair with $n = T_k(m)$ for a middle n	Lemma I.1.3
lattice point	a pair (m, k) with $m \equiv 7 \pmod{8}$ squarefree and k an odd multiple of m'	§1
wings (i), (ii)	kernels with $m \nmid U_1$, and with $m \mid U_1$	Corollary I.2.4
rank $\alpha_m(p)$	the smallest $k \geq 1$ with $p \mid T_k(m)$	Lemma I.4.1
witness	a prime that divides a number exactly once, which shows that the number is not powerful; the number is mostly $T_k(m)$ or M_d	Theorem I.4.6
class 7, class 3	kernels $m \equiv 7$, resp. $3 \pmod{8}$; they belong to middles $n \equiv 0$, resp. $2 \pmod{4}$	Lemma I.1.4
$\tau(k)$	the number of divisors of k	standard
$\text{rad}(N)$	the product of the distinct primes dividing N	standard
primitive part M_d , building block	the part of $T_d(m)$ on the primes of rank exactly d	Proposition I.6.8
Φ_n	the n -th cyclotomic polynomial	standard
Wieferich prime (to base 2)	a prime p with $p^2 \mid 2^{p-1} - 1$	standard
Wieferich prime for ε_m	an odd prime $p \nmid m$ with $p^2 \mid T_{\alpha_m(p)}(m)$	Proposition I.6.10
abc conjecture	$c < C_\epsilon \text{rad}(abc)^{1+\epsilon}$ for coprime $a + b = c$ and every $\epsilon > 0$	Remark I.4.12
certified prime	a prime proved by a deterministic test or by a certificate that a second program checks	Proposition I.6.8
ψ_{12}	the smallest number that passes the strong test to the first twelve prime bases without being prime, 318 665 857 834 031 151 167 461 [65]	Proposition I.6.8
V_n	the Lucas sequence $V_n(2T_1, 1) = 2T_n$	Proposition I.6.8
B_1	the first-stage bound of the elliptic curve method	Table 5

Table 2: Symbols and terms of Part I.

2 The height inequality

Theorem I.2.1 (height inequality). [proved] *Let $m \equiv 7 \pmod{8}$ be squarefree and let k be an odd multiple of m' , so that $m \mid U_k$. Then*

$$\log_{10} T_k(m) \geq m'(1.5 \log_{10} m - \log_{10} m') - \log_{10} 2.$$

In particular this bound holds for the middle $n = T_k \equiv 0 \pmod{4}$ of every pair of powerful numbers at distance two, and so for the middle of every triple of consecutive powerful numbers.

Proof. Put $d = m/m'$. By definition d divides U_1 , so $U_1 \geq d$. Hence $T_1 = \sqrt{mU_1^2 + 1} > \sqrt{m}U_1 \geq m^{3/2}/m'$. Since $T_k = (\varepsilon_m^k + \varepsilon_m^{-k})/2 \geq \varepsilon_m^k/2 \geq T_1^k/2$ and $k \geq m'$, we get $T_k \geq T_1^{m'}/2 > (m^{3/2}/m')^{m'}/2$. Take logarithms. $\square$

In words. The larger m' is, the larger the middle must be: its number of digits exceeds $m' \log_{10}(m^{1.5}/m') - 1$. This already forces the middle of any triple with a large m' to be enormous.

We did not find this inequality in the literature. Aktaş and Murty [1, proof of Theorem 1.3, p. 251] bound the index using only the trivial lower bound for the fundamental solution, and Chan [16, Section 2] counts solutions of Thue equations; neither uses that m' divides the index. Mollin and Walsh [45, p. 111] and Stephens and Williams [66, p. 619] note that k must be large.

Remark I.2.2 (the constant in the height bound). [proved] All terms of the binomial expansion $T_k = \sum_j \binom{k}{2j} T_1^{k-2j} (mU_1^2)^j$ are nonnegative, so $T_k \geq T_1^k$. Hence the term $-\log_{10} 2$ in Theorem I.2.1 can be dropped:

$$T_k \geq (m^{3/2}/m')^{m'}.$$

The proof uses only that m is odd and squarefree and that $m' \mid k$. So the same bound holds for kernels $m \equiv 3 \pmod{8}$, which carry the pairs with middle $\equiv 2 \pmod{4}$ (Lemma I.1.4).

In words. A slightly better bound holds, without the factor $1/2$, and it holds for both kinds of even middles.

Corollary I.2.3 (kernel bound). [proved] *Let m and k be as in Theorem I.2.1. If $m \nmid U_1$, then $T_k > m^{4.5}/27$. If $m \mid U_1$, then $T_k \geq T_1 > m^{3/2}$.*

Proof. If $m \nmid U_1$, then $m' > 1$, and m' is odd, so $m' \geq 3$. By Remark I.2.2, $\ln T_k \geq f(m')$ with $f(x) = x(1.5 \ln m - \ln x)$. Since $f'(x) = 1.5 \ln m - \ln(ex)$, the function f increases on $[3, m]$ as soon as $m^{3/2}/e > m$, that is, for $m \geq 8$. As $m' \leq m$, we get $\ln T_k \geq f(3) = 4.5 \ln m - 3 \ln 3$, which is the claim. The only kernel below 8 is $m = 7$, where $m' = 7$; there the smallest term is $T_7(7) \geq T_1(7)^7 = 8^7$, far above $7^{4.5}/27$. If $m \mid U_1$, then $U_1 \geq m$, so $T_1 > \sqrt{m}U_1 \geq m^{3/2}$, and $T_k \geq T_1$. $\square$

In words. If m does not divide U_1 , the middle is larger than $m^{4.5}/27$. If m divides U_1 , we only know that the middle is larger than $m^{1.5}$.

Corollary I.2.4 (two wings). [proved] *Let $H > 0$. Let $n < 10^H$ be the middle of a triple of consecutive powerful numbers, or the middle $n \equiv 0 \pmod{4}$ of a pair of powerful numbers at distance two, and let $m = \text{sqfree}(n^2 - 1)$. Then*

(i) *either $m \nmid U_1$ and $m < (27 \cdot 10^H)^{2/9}$,*

(ii) *or $m \mid U_1$ and $m < 10^{2H/3}$.*

Proof. By Lemmas I.1.1 and I.1.3, $n = T_k(m)$ for a lattice point (m, k) . In case $m \nmid U_1$, Corollary I.2.3 gives $m^{4.5}/27 < n < 10^H$; in case $m \mid U_1$ it gives $m^{3/2} < n < 10^H$. $\square$

In words. Every triple below 10^H has its kernel in one of two ranges. We call them wing (i), for kernels that do not divide U_1 , and wing (ii), for kernels that do. Wing (ii) allows much larger kernels.

The condition that separates the wings, $m \mid U_1$ or not, has an exact description by Wieferich primes in number fields, due to Fellini and Murty [26, Theorem 1.1]. We use only the different height behavior of the two wings.

Wing (ii) is the weak side. For a kernel with $m \mid U_1$ only the bound $n > m^{3/2}$ of Corollary I.2.3 is available. All such kernels up to $1.5 \cdot 10^{12}$ are known, and up to $5.325 \cdot 10^{13}$ if Reinhart's extended search missed nothing (Remark I.2.5); above that we know nothing about their size beyond Corollary I.2.3. A lower bound for $T_1(m)$ that grows faster with m in this class would close this gap, and we do not have one. Park [50] proves lower bounds for fundamental units that hold for almost all members of certain families. A statement about almost all members does not exclude a single exception, so it does not close the gap.

Remark I.2.5 (Reinhart's list). [literature] The squarefree m with $m \mid U_1(m)$ are the members of OEIS A135735 [49]. Reinhart proved by computer search that the list is complete up to $1.5 \cdot 10^{12}$ [52, Remark 5.5]. His later search covers the squarefree m up to $5.325 \cdot 10^{13}$ and found four further members, but he writes that he does not claim that these are all the members in that range, for lack of an independent double check, and that it is likely that all were found [53, Section 3]. We therefore keep two bounds apart: below $1.5 \cdot 10^{12}$ the list is complete; below $5.325 \cdot 10^{13}$ it is complete if the extended search missed nothing. Every statement that uses the larger bound says so. Of the members $m \equiv 7 \pmod{8}$, 3 lie below the smaller bound and 4 below the larger one.

In words. A published list of the dangerous kernels exists. Up to $1.5 \cdot 10^{12}$ its author proved it complete. Up to $5.325 \cdot 10^{13}$ he searched everything, but could not afford a second, independent run, and says so. We keep the two levels apart, and every statement below names the one it uses.

3 The reduction theorem

Theorem I.3.1 (reduction). [computed] *Among the 10 132 126 squarefree $m \equiv 7 \pmod{8}$ with $m \leq 10^8$, exactly 83 pass the discard criterion described below, and 79 of these have an odd multiple k of m' with $T_k(m) < 10^{2000}$. Together they have exactly 445 such indices k . For 74 of them T_k is odd. For each of the remaining 371 there is a prime $p < 2 \cdot 10^5$ that divides T_k exactly once. Consequently, there is no triple of consecutive powerful numbers whose middle n satisfies $n < 10^{2000}$ and $\text{sqfree}(n^2 - 1) \leq 10^8$.*

Proof. By Lemma I.1.1, such a triple is a lattice point (m, k) with $m \leq 10^8$ and $T_k < 10^{2000}$ at which T_k is even and powerful. An odd T_k fails the first condition. A prime that divides T_k exactly once shows that T_k is not powerful. It remains to check that the list of points is complete, which is the computation below. $\square$

In words. We checked every possible address with a kernel up to 10^8 and a middle below 10^{2000} . At each of the 445 addresses, either the middle is odd, or a small prime divides it only once. In both cases the middle is not powerful, so no triple lives there. The theorem says nothing about kernels above 10^8 or middles above 10^{2000} .

The computation. Put $H = 2000$. For each $m \leq 10^8$ the continued fraction of $\sqrt{m}$ is computed twice: modulo m , which gives $U_1 \bmod m$ and hence m' by Lemma I.1.2, and in scaled floating point, which gives $\log_{10} T_1$; on a sample of 3531 kernels, all up to $2 \cdot 10^4$ and the others drawn at random up to 10^8 , its relative error against the exact value is below $3 \cdot 10^{-16}$. A kernel is discarded when $m' > (H + 0.31)/\log_{10} T_1 + 1$. This is rigorous: for every admissible k , $\log_{10} T_k \geq k \log_{10} T_1 - 0.31 \geq m' \log_{10} T_1 - 0.31$, so a discarded kernel has no index with $T_k < 10^H$; the extra 1 is a margin against rounding. It absorbs any relative error in $\log_{10} T_1$ up to $\log_{10} T_1/(H + 0.31)$, and since $T_1 \geq 4$ for every $m \equiv 7 \pmod{8}$, an error below $3 \cdot 10^{-4}$ is enough, and the measured error is far below that. For the 83 surviving kernels the unit is computed exactly and compared with the modular route, and the run stops on any mismatch. Each $T_k < 10^H$ is computed exactly and tested for parity and for a prime $p < 2 \cdot 10^5$ with $v_p(T_k) = 1$. The discard criterion was also rechecked without floating point, from the bound $\log_{10} T_1 \geq (\ell - 1) \log_{10} \varphi - \log_{10} \sqrt{5}$, where ℓ is the period length of the continued fraction and φ the golden ratio; the numerators of the convergents grow at least like Fibonacci numbers. This was done for a random sample of 530 discarded kernels up to 10^8 and for every discarded kernel up to $2 \cdot 10^5$: none is contradicted, and the smallest margin by which a discarded kernel exceeds 10^H is 21.5 orders of magnitude. Three versions of the scan, which differ in speed and not in the criterion, give identical lists of points and witnesses. The list of points with their witnesses is part of the data deposit. Figure 3 shows the numbers.

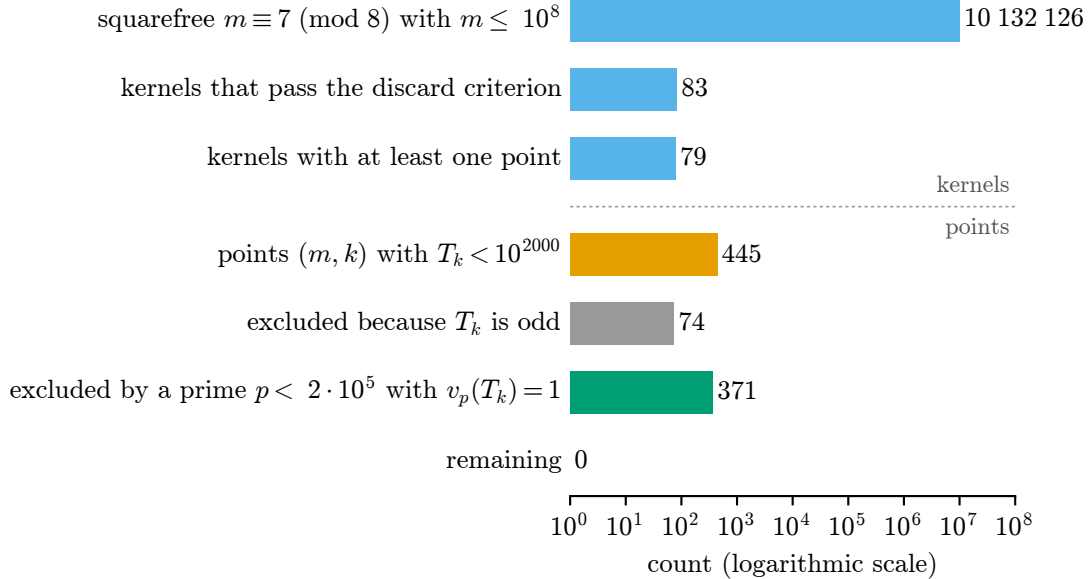


Figure 3: **No point of the box remains.** From all kernels to no candidate (Theorem I.3.1). The horizontal axis is logarithmic. The upper three bars count kernels, the lower ones count points (m, k) ; the two exclusions add up to all points, and none remains.

An example. Of the 445 points, the one with the smallest middle is $(m, k) = (7, 7)$, the example of Mollin and Walsh [45, p. 111]. There $\varepsilon_7 = 8 + 3\sqrt{7}$, so $7 \nmid U_1$ and $m' = m$, and

$$n = T_7(7) = 130\,576\,328 = 2^3 \cdot 29 \cdot 197 \cdot 2857.$$

Since $n \equiv 0 \pmod{4}$ and m' divides k , Lemma I.1.3 shows that $n - 1$ and $n + 1$ are powerful. But the prime 29 divides n exactly once, so n is not powerful, and $(n - 1, n, n + 1)$ is not a triple. The prime 29 is the witness recorded for this point.

Remark I.3.2 (weaker bounds with less work). [proved] The same method gives weaker bounds with less work. The list OEIS A135735 contains all squarefree m with $m \mid U_1(m)$ up to $1.5 \cdot 10^{12}$, and up to $5.325 \cdot 10^{13}$ if Reinhart's extended search missed nothing (Remark I.2.5). A kernel with $m \mid U_1$ that is not on this list is therefore larger than $1.5 \cdot 10^{12}$, and Corollary I.2.3 gives $n > m^{3/2} > 1.83 \cdot 10^{18}$ for every triple with such a kernel; under the assumption on the extended search the bound is $3.88 \cdot 10^{20}$. The enumeration of consecutive powerful pairs behind the b-file of OEIS A060355 is described by Mian and Siddique as exhaustive below 10^{22} but not certified [48, 43]; since every triple contains two such pairs, it excludes triples below that bound on the same terms. The bound above is stated as Proposition I.6.2(d); here it only illustrates the method.

In words. A short list plus the height bound already pushes some triples beyond $1.83 \cdot 10^{18}$. This is only an illustration: other searches reach further, and the reduction theorem is the actual result of this section.

4 The exponent condition and the witness reduction

For a prime p that does not divide m , this section describes at which indices k it divides T_k , and with which exponent. This turns the question whether T_k is powerful into a question about finitely many primes.

Lemma I.4.1 (Bennett and Walsh [8, proof of Lemma 3.3, p. 3484]; Lehmer [35]). [literature] *Let p be a prime with $p \nmid m$. Either p divides no T_k , or there is a smallest index $\alpha_m(p) \geq 1$ with $p \mid T_{\alpha_m(p)}$. In the second case, $p \mid T_k$ if and only if k is an odd multiple of $\alpha_m(p)$.*

We call $\alpha_m(p)$ the *rank* of p . Bennett and Walsh state the lemma in the proof of their Lemma 3.3 and characterize the rank in their Lemma 5.1 [8, p. 3486], where they call it quite standard and refer to Lehmer.

In words. A prime appears in the sequence $T_1, T_2, T_3, \dots$ for the first time at its rank α . After that it divides exactly the terms $T_\alpha, T_{3\alpha}, T_{5\alpha}, \dots$, and no others.

Lemma I.4.2 (Bennett and Walsh [8, Theorem 1.2]). [literature] *Let m and $b > 1$ be squarefree. There is at most one index k with $T_k = bx^2$ for an integer x , and if it exists, then $k = \alpha_m(b)$, the smallest k with $b \mid T_k$.*

By Lemma I.4.1, $\alpha_m(b)$ exists if and only if every prime of b has a rank and all these ranks have the same 2-adic valuation; then $\alpha_m(b)$ is their least common multiple.

In words. A term T_k can be b times a square for at most one index k , and this index is the first place where b divides the sequence.

Lemma I.4.3 (valuation identity). [proved] *Let p be an odd prime with $p \nmid m$ and $p \mid T_r$. Then $v_p(T_{rt}) = v_p(T_r) + v_p(t)$ for every odd t .*

Proof. Here $v_p(z)$, for $z \in \mathbb{Z}[\sqrt{m}]$, is the largest e with $p^e \mid z$ in $\mathbb{Z}[\sqrt{m}]$, the least p -adic valuation of the two coefficients of z . It is not multiplicative when p splits, but it does not change under multiplication by a unit or by an element congruent to 1 modulo p , and $v_p(tz) = v_p(t) + v_p(z)$ for $t \in \mathbb{Z}$; this is all that the standard proof of the lifting-the-exponent lemma uses. Put $w = \varepsilon_m^{2r}$ and $u = w + 1 = \varepsilon_m^r \cdot 2T_r$, so $v_p(u) = v_p(T_r) \geq 1$. For odd t ,

$$2T_{rt} = \varepsilon_m^{-rt} (w^t + 1) = \varepsilon_m^{-rt} (1 - (1 - u)^t),$$

and the lifting-the-exponent lemma in $\mathbb{Z}[\sqrt{m}]$ gives $v_p(1 - (1 - u)^t) = v_p(u) + v_p(t)$. It applies because p is odd, $p \nmid m$, and p divides u but not $1 - u$; the proof is the same as for integers. $\square$

In words. If a prime p divides T_r , then in T_{rt} it appears exactly as often as in T_r , plus as often as it divides t . So a prime that appears once in T_r appears once in T_{rt} , unless it also divides t .

Proposition I.4.4 (exponent-3 condition). [proved] *Let T_k be an even powerful number with k odd and $m' \mid k$, and let $b = \text{sqfree}(T_k)$, so that $T_k = b^3 z^2$ for an integer z . If $b > 1$, then $k = \alpha_m(b)$, and for every odd prime $p \mid b$*

$$v_p(T_{\alpha_m(p)}) + v_p(\alpha_m(b)/\alpha_m(p)) \geq 3.$$

Proof. We have $T_k = b(bz)^2$, so $k = \alpha_m(b)$ by Lemma I.4.2. Now let $p \mid b$ be odd. Then $p \mid T_k$, so $k = \alpha_m(p)t$ with t odd by Lemma I.4.1, and $v_p(T_k) = v_p(T_{\alpha_m(p)}) + v_p(t)$ by Lemma I.4.3. As p divides the powerful number T_k to an odd power, this power is at least 3. $\square$

For general Lucas sequences the valuation identity is standard; see Sanna [58]. Ballot [5, p. 265, (2)] printed it with the term $(\nu - 1) + v_p(n)$, and the errata [4] corrects this to $\nu + v_p(n/\rho)$. The corrected form is the one in Lemma I.4.3. On 56 cases of our sequence, the corrected form holds in all of them and the printed form in 0; on 50 cases of the Fibonacci numbers, the corrected form holds in all of them.

In words. If the middle T_k is powerful, then every prime that divides it an odd number of times divides it at least three times. This gives a condition on the first term $T_{\alpha(p)}$ in which the prime appears. Proposition I.4.4 does not prove the conjecture of Mollin and Walsh [45, p. 111] that T_k is never an even powerful number for odd $k > 1$; it says what a counterexample would have to satisfy.

Theorem I.4.5 (the b -box). [computed] *There is no triple of consecutive powerful numbers whose middle n has kernel $m \leq 10^6$ and $\text{sqfree}(n) \leq 10^4$, whatever the size of n .*

Proof. If $\text{sqfree}(n) = 1$, then n is a square, and Theorem I.5.4 gives $m \mid U_1$; no such kernel with T_1 even lies below 10^6 : the kernels $m \mid U_1$ below this bound are listed completely [52, Remark 5.5], and the only one with $m \equiv 7 \pmod{8}$, 209 991, has T_1 odd (Theorem I.5.4; OEIS [49]). Otherwise let $b = \text{sqfree}(n) > 1$ and $n = T_k(m)$. By Proposition I.4.4, $k = \alpha_m(b)$, so $\alpha_m(b)$ must exist, be odd and be a multiple of m' , and b^3 must divide $T_{\alpha_m(b)}$. We checked all pairs. There are 101 329 squarefree $m \equiv 7 \pmod{8}$ with $m \leq 10^6$; for 32 118 of them T_1 is

odd, so they carry no even middle. The others give 421 010 513 pairs (m, b) with squarefree $b \leq 10^4$, $b = 1$ included. In 343 504 538 pairs some prime of b has no rank, in 16 603 302 the ranks have different 2-adic valuations, in 11 242 224 the index $\alpha_m(b)$ is even, and in 49 659 524 it is not a multiple of m' ; this class contains the pairs with $b = 1$, where $k = 1$ while $m' > 1$ for every such kernel with T_1 even [53]. For each of the remaining 925 pairs, b^3 does not divide $T_{\alpha_m(b)}$; this is tested modulo b^3 . $\square$

In words. If the kernel is at most 10^6 and the part of the middle with odd exponents is at most 10^4 , there is no triple, however large the middle is. This result has no height limit, unlike Theorem I.3.1.

Theorem I.4.6 (reduction to a witness statement). [proved] *Let $\mathcal{L}$ be the set of lattice points (m, k) with $T_1(m)$ even. For $(m, k) \in \mathcal{L}$ one has $4 \mid T_k(m)$ and $\gcd(T_k(m), m) = 1$. The following statements are equivalent.*

- (i) *There are no three consecutive powerful numbers.*
- (ii) *For every $(m, k) \in \mathcal{L}$ there is a prime p with $v_p(T_k(m)) = 1$, a witness.*
- (iii) *For every $(m, k) \in \mathcal{L}$ there is an odd prime $p \nmid m$ such that k is an odd multiple of $\alpha_m(p)$, $v_p(T_{\alpha_m(p)}(m)) = 1$, and $p \nmid k/\alpha_m(p)$.*

In particular, if $T_k(m)$ is powerful, then $p^2 \mid T_k(m)$ for every prime p with $\alpha_m(p) = k$.

Proof. By Lemmas I.1.1 and I.1.3, a triple exists if and only if $T_k(m)$ is powerful for some $(m, k) \in \mathcal{L}$. For every $(m, k) \in \mathcal{L}$ the number T_k is even (proof of Lemma I.1.4), $U_k = mw$ by Lemma I.1.2 with w odd, and $T_k^2 = 1 + m^3 w^2 \equiv 1 + 7 \equiv 0 \pmod{8}$, because $m \equiv 7 \pmod{8}$; hence $4 \mid T_k$. From $T_k^2 \equiv 1 \pmod{m}$ we get $\gcd(T_k, m) = 1$. So T_k is powerful if and only if no odd prime $p \nmid m$ has $v_p(T_k) = 1$, which proves (i) $\Leftrightarrow$ (ii). For (ii) $\Leftrightarrow$ (iii): by Lemma I.4.1 an odd prime $p \nmid m$ divides T_k exactly when k is an odd multiple of $\alpha_m(p)$, and then $v_p(T_k) = v_p(T_{\alpha_m(p)}) + v_p(k/\alpha_m(p))$ by Lemma I.4.3. So $v_p(T_k) = 1$ exactly when $v_p(T_{\alpha_m(p)}) = 1$ and $p \nmid k/\alpha_m(p)$. For $\alpha_m(p) = k$ the identity reads $v_p(T_k) = v_p(T_{\alpha_m(p)})$, which gives the last claim. $\square$

In words. Three consecutive powerful numbers do not exist exactly when every address has a witness: a prime that divides the middle only once. The last sentence says what a counterexample would need. Every prime that appears in T_k for the first time would have to divide it at least twice, a coincidence of the kind known as a Wieferich condition.

The reduction is due to Mollin and Walsh, who state it and conjecture its conclusion [45, p. 111]. The mechanism has a published general form: Fellini and Murty [26, Lemma 5.6] split $\alpha^n - 1$ into a squarefree and a powerful part in the ring of integers of a number field and show that every prime of the squarefree part is a base- α non-Wieferich prime. Kym [33, Corollary 6.6 and Remark 6.7] gives the closest published form for Lucas sequences: primitive prime divisors in a coprime product must appear with exponent at least 2. Kym studies a different equation and not consecutive powerful numbers. What is specific here is the translation into the addresses (m, k) and the finite bookkeeping of Section 6.

Lemma I.4.7 (the T_1 -part of a lattice point). [proved] *Let m be odd and squarefree with $T_1 = T_1(m)$ even, and let k be odd. Then $T_1 \mid T_k$ and T_k/T_1 is odd, so $v_2(T_k) = v_2(T_1)$. For every odd prime $p \mid T_1$ one has $\alpha_m(p) = 1$ and $v_p(T_k) = v_p(T_1) + v_p(k)$. Consequently:*

- (a) *the 2-adic valuation of the middle of any triple with kernel m is $v_2(T_1(m))$, which is at least 2 for $m \equiv 7 \pmod{8}$;*
- (b) *if T_k is powerful, then k is divisible by every odd prime p with $v_p(T_1) = 1$.*

Proof. We have $T_k = \sum_{j=0}^{(k-1)/2} \binom{k}{2j} T_1^{k-2j} (mU_1^2)^j$, and every term contains T_1^{k-2j} with $k-2j \geq 1$, so $T_1 \mid T_k$. In T_k/T_1 the term with $j = (k-1)/2$ equals $k(mU_1^2)^{(k-1)/2}$, which is odd because k, m and U_1 are odd; U_1 is odd since T_1 is even and $T_1^2 - mU_1^2 = 1$. Every other term contains T_1^{k-2j-1} with an even exponent at least 2 and is even. So T_k/T_1 is odd. The p -adic formula is the valuation identity of Lemma I.4.3 with $\alpha_m(p) = 1$. Part (a) follows, where $v_2(T_1) \geq 2$ for $m \equiv 7 \pmod{8}$ because $T_1^2 = 1 + mU_1^2 \equiv 1 + 7 \equiv 0 \pmod{8}$, and part (b) follows because $v_p(T_k) = 1 + v_p(k) \geq 2$ forces $p \mid k$. $\square$

In words. All terms T_k with odd k contain T_1 as a factor, with the same power of 2. Every odd prime that divides T_1 only once must also divide the index k of a powerful middle. For $m = 7$ it gives nothing, because $T_1 = 8$; the ladder of Theorem I.5.1 applies the same idea at the index $k_0 = m'$ instead of at 1, and for $m = 7$ its first step is the one of Mollin and Walsh.

Corollary I.4.8 (the divisor count of the Wieferich conditions). [proved] *Let $m > 1$ be squarefree and let k be odd. For every divisor $d \geq 5$ of k there is a prime p_d with $\alpha_m(p_d) = d$, and:*

- (i) *every odd prime $p \nmid m$ with $\alpha_m(p) = d$ satisfies $p \equiv \pm 1 \pmod{4d}$, so $p \geq 4d - 1$; in particular no prime dividing d has rank d ;*
- (ii) *distinct divisors d give distinct primes p_d , and each p_d divides $T_k(m)$; hence the number $\omega(T_k(m))$ of distinct prime factors satisfies $\omega(T_k(m)) \geq \#\{d \mid k : d \geq 5\}$, which is $\tau(k) - 1$ if $3 \nmid k$ and $\tau(k) - 2$ if $3 \mid k$;*
- (iii) *if $T_k(m)$ is powerful, then for every divisor $d \geq 5$ of k and every prime p with $\alpha_m(p) = d$, in particular for p_d , either $p^2 \mid T_d(m)$ or $p \mid k/d$; the second case forces $k \geq d(4d - 1)$.*

Proof. Put $\alpha = T_1 + U_1\sqrt{m}$ and $\beta = T_1 - U_1\sqrt{m}$. They are the distinct real roots of $z^2 - 2T_1z + 1$, so $D_n = (\alpha^n - \beta^n)/(\alpha - \beta) = U_n/U_1$ is a Lucas sequence in the sense of Carmichael, and $D_{2n} = D_n \cdot 2T_n$. *Existence.* Let $d \geq 5$ divide k ; then d is odd and $2d \geq 10$. By Carmichael's theorem [15], in the form of Yabuta [73, Theorem 1], the term D_{2d} has a primitive divisor p : $p \mid D_{2d}$ and $p \nmid D_j$ for $0 < j < 2d$. Since $D_{2d} = D_d \cdot 2T_d$ and $p \nmid D_d$, we get $p \mid 2T_d$; since $p \nmid D_2 = 2T_1$, the prime p is odd, so $p \mid T_d$. If p divided some T_j with $j < d$, then $p \mid 2T_j \mid D_{2j}$ with $2j < 2d$, which is impossible; so $\alpha_m(p) = d$. Also $p \nmid m$, because $p \mid m$ would give $T_j^2 = 1 + mU_j^2 \equiv 1 \pmod{p}$ for all j . (i) $p \mid T_d$ means $\varepsilon_m^{2d} \equiv -1 \pmod{p}$, and d is the smallest index with this property, so ε_m has order $4d$ modulo p . The elements of norm one in $(\mathbb{F}_p[\sqrt{m}])^\times$ form a cyclic group of order $p - (\frac{m}{p})$, and ε_m has norm one, so $4d \mid p - (\frac{m}{p})$ and $p \equiv \pm 1 \pmod{4d}$. This uses only $\alpha_m(p) = d$, so it holds for every prime of rank d . As $p \geq 4d - 1 > d$, no prime dividing d has rank d . (ii) The rank determines d , so distinct d give

distinct primes. Since k/d is odd, Lemma I.4.1 gives $p_d \mid T_k$. The odd divisors of k below 5 are 1 and 3. (iii) Let p be any prime with $\alpha_m(p) = d$. As above $p \nmid m$, and $p \mid T_k$ as in (ii). If T_k is powerful, then $v_p(T_k) \geq 2$, and $v_p(T_k) = v_p(T_d) + v_p(k/d)$ by Lemma I.4.3. So $p^2 \mid T_d$ or $p \mid k/d$; in the second case $k/d \geq p \geq 4d - 1$ by (i). The prime p_d is one such prime. $\square$

In words. Every divisor $d \geq 5$ of k brings its own new prime into T_k . For T_k to be powerful, each of these primes must pass a Wieferich-type test, unless it also divides k/d , which forces $k \geq d(4d - 1)$ and is therefore impossible when d is large compared with $\sqrt{k}$. So a counterexample needs many coincidences at once, each at a different prime; and every further prime of the same rank d must pass the same test. When k is a prime, the count gives only a single condition.

The argument is standard, and it is in print for the Fibonacci and Lucas numbers. The earliest occurrence we found is Luca [38, p. 267], who derives $\omega(F_n) \geq \tau(n) - 2$ from Carmichael's theorem in the same way. Bugeaud, Luca, Mignotte and Siksek [14, proof of Theorem 5] use the same step. Pongsriiam [51, Lemmas 2.5(i) and 3.4(i)] states it, including the version for the companion sequence used above, and Somer and Křížek [64, Lemma 2.7] collect the general facts for Lucas sequences, with the congruence modulo $2n$. What we did not find in the literature is the statement for the norm-one Pell sequences and the sharpening from modulus $2d$ to modulus $4d$, which follows from $N(\varepsilon_m) = 1$; neither is a new mechanism. Mollin and Walsh [45, p. 111] already point to the theory of Lucas and Lehmer [39, 35] for this step.

Theorem I.4.9 (Granville [29, Theorem, p. 216]). [literature] *If there are no three consecutive powerful numbers, then there are infinitely many primes p with $p^2 \nmid 2^p - 2$.*

In words. If triples never exist, then infinitely many primes are not Wieferich primes to base 2. Nobody knows how to prove this last statement without extra assumptions. So a proof that triples never exist would also settle an open problem about Wieferich primes.

That infinitely many primes are non-Wieferich is known under the *abc* conjecture. Silverman [62, Theorem 1] proved it for a fixed rational base; Kotyada and Muthukrishnan [32, Theorem 1.1] for the fundamental unit of a real quadratic field of class number one; and Fellini and Murty [26, Theorem 1.2] for number fields, under the *abc* conjecture of Masser. Fellini and Murty also derive it from the finiteness of base- α super-Wieferich primes [26, Theorem 1.3]; in the rational case this is Theorem 1.4 of Murty and Séguin [46]. All of these split $\alpha^n - 1$ into a squarefree and a powerful part, as this section does. The same splitting, applied to the terms of a Lucas sequence, shows under the *abc* conjecture that there are only finitely many powerful terms: Ribenboim and Walsh [54, Corollary 2], and Yabuta [74, Theorem 2.3], who uses the primitive divisor theorem of Bilu, Hanrot and Voutier [9]. For the sequences $T_k(m)$ this gives, under *abc*, finitely many k for each fixed m . The *abc* conjecture also implies that there are only finitely many triples [1, Section 6], [69, Section 1]. None of these conditional results is effective.

Granville's proof is a construction [29, pp. 216–217]. If every prime $p > p_0$ satisfied $p^2 \mid 2^p - 2$, it would produce a triple whose middle is a power of 2, and in fact one with middle A^n for every $n \geq 1$, where A is the number of his construction, so that the theorem already holds if there are only finitely many triples; Granville states it under the conjecture of Mollin and Walsh. In our notation such a middle is $T_k(m) = 2^a$. By Lemma I.4.7 the number T_1 divides 2^a and T_k/T_1 is odd, so $T_k = T_1$ and $k = 1$. Proposition I.6.7 treats this case numerically.

Remark I.4.10 (what the witness condition does not follow from). [proved] The witness condition does not follow from coprimality. If n and $n+1$ are powerful, then $\gcd(n, n+2) \mid 2$ and $\gcd(n+1, n+2) = 1$, so every odd prime factor of $n+2$ divides neither n nor $n+1$. This says which primes occur in $n+2$, but nothing about their exponents. An example is $n = 9800$: both n and $n+1$ are powerful, and $n+2 = 2 \cdot 13^2 \cdot 29$, so the new prime 13 appears twice. If n is even and powerful, then $4 \mid n$, so $n+2 \equiv 2 \pmod{4}$ and $v_2(n+2) = 1$; this is the reason for the parity filter of Theorem I.3.1, which keeps only kernels with T_1 even. The difficulty lies entirely in the case of odd n , where the witness must be odd. The step from “a new prime divides $n+2$ ” to “some prime divides $n+2$ exactly once” is equivalent to the non-existence of a triple, and so, by Theorem I.4.6, to the witness statement (ii) there, which asks for the witness in the middle of a triple instead of in its outer number. An argument of this shape has been posted [11]; this remark concerns that step and is not an assessment of the preprint.

In words. Knowing that a prime is new in $n+2$ does not tell us how often it appears there. The pair 9800, 9800+1 shows that a new prime can appear twice in the next number.

Remark I.4.11 (where a witness in an outer number can sit). [proved] Let m be odd and squarefree with $T_1 = T_1(m)$ even, let k be odd, and let γ be a square root of $\varepsilon = T_1 + U_1\sqrt{m}$ in an extension field, so that γ^{-1} is a square root of ε^{-1} . Then $(\gamma \pm \gamma^{-1})^2 = \varepsilon + \varepsilon^{-1} \pm 2 = 2(T_1 \pm 1)$, and in the same way $(\gamma^k \pm \gamma^{-k})^2 = 2(T_k \pm 1)$. The quotients

$$V_k = \frac{\gamma^k + \gamma^{-k}}{\gamma + \gamma^{-1}} = \sum_{j=0}^{k-1} (-1)^j \gamma^{k-1-2j}, \quad W_k = \frac{\gamma^k - \gamma^{-k}}{\gamma - \gamma^{-1}} = \sum_{j=0}^{k-1} \gamma^{k-1-2j}$$

are polynomials with integer coefficients in $(\gamma + \gamma^{-1})^2 = 2(T_1 + 1)$: the terms j and $k-1-j$ carry the same sign and combine to $\gamma^r + \gamma^{-r}$ with r even, which is an integer polynomial in $\gamma^2 + \gamma^{-2} = (\gamma + \gamma^{-1})^2 - 2$. Hence V_k and W_k are integers, and squaring gives

$$T_k + 1 = (T_1 + 1) V_k^2, \quad T_k - 1 = (T_1 - 1) W_k^2.$$

Reducing modulo a prime $p \mid T_1 + 1$ amounts to setting $(\gamma + \gamma^{-1})^2 = 0$, that is $\gamma^2 = -1$, where $V_k = k\gamma^{k-1} = \pm k$; reducing modulo a prime $p \mid T_1 - 1$ amounts to $(\gamma - \gamma^{-1})^2 = 0$, that is $\gamma^2 = 1$, where $W_k = k$. So $V_k \equiv \pm k \pmod{p}$ for every prime $p \mid T_1 + 1$, and $W_k \equiv k \pmod{p}$ for every prime $p \mid T_1 - 1$; all these primes are odd because T_1 is even.

Now let $n+1 = T_k(m)$, so that n and $n+2$ are the outer numbers. Every prime that divides n or $n+2$ but not $T_1 \mp 1$ appears there with an even exponent, so the squarefree parts of the outer numbers are those of $T_1 - 1$ and $T_1 + 1$, whatever k is. An odd prime q divides $n+2$ exactly once if and only if q divides $T_1 + 1$ exactly once and $q \nmid k$, and likewise for n and $T_1 - 1$; if $q \mid k$, then $q \mid V_k$ and the exponent is at least 3. Such a prime divides m and not U_1 , because $(T_1 - 1)(T_1 + 1) = mU_1^2$ and q divides exactly one of the two coprime factors, to an odd power; so it divides m' . This settles whether the structure forces an odd witness in $n+2$ as it forces the prime 2 when n is even: the candidates are exactly the odd primes that divide $T_1(m) + 1$ exactly once, each of them is a witness unless it divides k , and on a lattice point, where $m' \mid k$ (Lemma I.1.3), every candidate divides k . If $T_1 + 1$ is a square there is no candidate at all, and $n+2$ is powerful for every odd k . This is the case for every prime kernel $m \equiv 7 \pmod{8}$, by an elementary argument: if T_1 were odd, then U_1 would be even and the

coprime numbers $(T_1 - 1)/2$ and $(T_1 + 1)/2$ would have product $m(U_1/2)^2$, so one of them would be a square a^2 and the other mb^2 with $b \geq 1$; the case $a^2 - mb^2 = -1$ is impossible for $m \equiv 3 \pmod{4}$, and $a^2 - mb^2 = 1$ would be a solution smaller than the fundamental one. So T_1 is even, $T_1 - 1$ and $T_1 + 1$ are coprime with product mU_1^2 , one of them is a square, and it is not $T_1 - 1 \equiv 3 \pmod{4}$. Hence no odd prime is forced in the way the prime 2 is, and for the outer numbers the question returns to Theorem I.4.6, which places the witness in the middle.

In words. The two neighbors of T_k inherit the primes of odd exponent from the two neighbors of T_1 ; everything else they gain comes in squares. So an odd witness in an outer number can only be a prime of $T_1 \pm 1$, and on a lattice point none of these primes is available.

Remark I.4.12 (two elementary constraints, and the size of the gap). [proved] First, if a prime q divides N and $q > N^{1/2}$, then $v_q(N) = 1$. So a powerful number has no prime factor above its square root, and a large prime factor is always a witness. Second, $n^2 - (n-1)(n+1) = 1$ is an *abc* equation. For a triple, all three numbers are powerful, so the radical of $(n-1)n(n+1)$ is at most $((n-1)n(n+1))^{1/2} < n^{3/2}$, while the largest term is n^2 . Hence an *abc* inequality with exponent below $4/3$ would exclude all large triples. (The *abc* conjecture states that for every $\epsilon > 0$ there is a constant C_ϵ with $c < C_\epsilon \text{rad}(abc)^{1+\epsilon}$ for all coprime positive integers with $a + b = c$, where rad is the product of the distinct prime factors.) The exponents derived from Baker's explicit form of the conjecture are larger than $4/3$: $7/4$ [34, Theorem 1], 1.72 [19, Theorem 1.4], and 1.7 [18, Theorem 4.1]; all three assume that explicit conjecture. The explicit form does give a finite bound: under it, Chim, Shorey and Sinha [19, Theorem 1.8] show that the middle of every triple is below 10^{51075} , and they cite a bound 10^{20000} attributed to Trudgian. Without any hypothesis, the best known bounds in the *abc* problem are exponential in a power of the radical [67].

In words. A very large prime factor is always a witness. The *abc* conjecture in a strong enough form would rule out all large triples. The explicit forms that have been worked out are themselves conjectures, and they give exponents too large for this. What is proved without any assumption is much weaker still.

Remark I.4.13 (the cyclotomic axis). [proved] If one member of a triple is a proper power x^r with $r \geq 2$, its neighbors are $x^r \pm 1$, and these split into values $\Phi_e(\pm x)$ of cyclotomic polynomials. Each factor gives further conditions on the possible witnesses. These conditions apply only when a member is a proper power, and they are treated in Part III. For $r = 2$, the number $x^2 + 1$ is powerful exactly when $x^2 - b^3y^2 = -1$ has a solution with b squarefree; this equivalence is due to De Koninck, Doyon and Luca [22, (1)–(2)].

In words. If one number of a triple is a square, a cube or a higher power, its neighbors split into smaller factors, and each factor brings its own conditions on a witness. Part III follows this route.

For a middle that is a cube, four shapes of the neighbors have been excluded, with primes p and q : Chan [17] excludes $x^3 - 1 = p^3y^2$ together with $x^3 + 1 = q^3z^2$, She [61] excludes $x^3 - 1 = p^2a^3$ together with $x^3 + 1 = q^2b^3$, and [59, Theorem 1] excludes the two mixed shapes, $x^3 - 1 = p^2a^3$ with $x^3 + 1 = q^3z^2$ and $x^3 - 1 = p^3y^2$ with $x^3 + 1 = q^2b^3$. That paper reduces both mixed shapes to the single equation $t^6 + t^3 + 1 = 3w^2$, whose rational solutions it determines through an elliptic quotient of rank 0. Each of these results excludes infinitely many candidate middles at once, by an equation and not by a computation up to a bound.

5 Exclusion regions and square-type middles

Theorem I.3.1 covers kernels up to 10^8 and middles below 10^{2000} . This section adds three results of a different shape: a strip of small kernels where every possible middle is huge, a box of larger kernels built from small primes, and a restriction for middles of a special form.

Theorem I.5.1 (the strip). [computed] *There are 206 squarefree $m \equiv 7 \pmod{8}$ with $m \leq 2000$. For 52 of them T_1 is odd, and no triple has such a kernel. For each of the other 154, the middle $n = T_k(m)$ of a triple with kernel m would have more than $8.4 \cdot 10^7$ decimal digits. For $m = 7$ it would have more than $10^{17.5}$ decimal digits.*

Proof. If T_1 is odd, then T_k is odd for every odd k (proof of Lemma I.1.4), so T_k is not the middle of a triple. For the other kernels we climb a ladder of two steps. It rests on Lemma I.4.1 and Lemma I.4.3: if a prime p divides T_{k_0} exactly once and k is an odd multiple of k_0 , then $v_p(T_k) = 1 + v_p(k/k_0)$, so T_k can be powerful only if p divides k/k_0 . Start with the smallest admissible index $k_0 = m'$. The primes that divide T_{k_0} exactly once force k to be a multiple of a larger index k_1 , and the same argument at k_1 gives an index k_2 . So $T_k \geq T_{k_2}$ whenever T_k can be powerful, and T_{k_2} has the stated number of digits. In the first run such primes were found at both steps for all but 14 kernels; a deeper search found them for these 14 as well, but for 2 of them only at the first step; there the bound from the first step is already larger than the smallest one. All other bounds are larger than the smallest one too, which occurs at $m = 223$. $\square$

For $m = 7$ the first step is due to Mollin and Walsh [45, p. 111]: the primes 29, 197 and 2857 divide $T_7(7)$ exactly once, so k must be a multiple of $7 \cdot 29 \cdot 197 \cdot 2857$.

In words. Of the kernels up to 2000, the 52 with T_1 odd carry no triple at all. For the other 154 a triple is not ruled out, but its middle would be gigantic: it would have more than $8.4 \cdot 10^7$ digits. For $m = 7$ even the number of its digits would be larger than $10^{17.5}$.

Remark I.5.2 (the bound of Chim, Shorey and Sinha). [literature] Chim, Shorey and Sinha [19, p. 440] state the lower bound $10^{3 \cdot 10^{13}}$ for the middle when $m \in \{15, 23, 31, 39, 47, 55, 87\}$, following the arguments of Mollin and Walsh, and do not give the computation. For $m = 15$, 31 and 87 the two-step ladder gives a larger bound, for $m = 23$ and 47 a smaller one. For $m = 39$ and 55 the number T_1 is odd, so these kernels carry no triple at all.

In words. For the kernels named by Chim, Shorey and Sinha, our ladder gives a larger bound for three and a smaller one for two; two of them carry no triple at all, because T_1 is odd.

Theorem I.5.3 (smooth kernels). [computed] *No triple of consecutive powerful numbers with middle $n < 10^{2000}$ has a kernel m with $10^8 < m \leq 10^{15}$ whose prime factors are all at most 100 and which has at least 4 prime factors. There are 889 437 such squarefree $m \equiv 7 \pmod{8}$. For 356 888 of them the period ℓ of the continued fraction of $\sqrt{m}$ exceeds 9573, which forces $T_1 > 10^{2000}$. None of the remaining 532 549 has an admissible index k with $T_k(m) < 10^{2000}$.*

Proof. The numerators A_j of the convergents of $\sqrt{m}$ satisfy $A_j \geq F_{j+2}$, where F_j are the Fibonacci numbers, and T_1 is the numerator at the end of the first period; the period is even because $x^2 - my^2 = -1$ has no integer solution for $m \equiv 3 \pmod{4}$. So $T_1 \geq F_{\ell+1} \geq \varphi^{\ell-1}$, and $\ell > 9573$ gives $T_1 > 10^{2000}$. For the other kernels the computation of Theorem I.3.1 finds no admissible index below 10^{2000} . $\square$

In words. Theorem I.3.1 stops at kernels up to 10^8 . For one kind of larger kernel, up to 10^{15} and with all prime factors at most 100, we can still rule out every triple with a middle below 10^{2000} . (Such a kernel above 10^8 automatically has at least 4 prime factors, since any 4 primes up to 100 have a product below 10^8 .)

Theorem I.5.4 (square-type middles). [computed] *Let $(n-1, n, n+1)$ be a triple of consecutive powerful numbers with $n = T_k(m)$. If n is a square or twice a square, in particular if n is a fourth power, then $k = 1$ and $m \mid U_1$. Hence either $m > 1.5 \cdot 10^{12}$, or m is one of 209 991, 4 099 215 and 117 477 414 815; if Reinhart's extended search missed nothing (Remark I.2.5), either $m > 5.325 \cdot 10^{13}$, or m is one of these three or 39 028 039 587 479. None of the four carries a triple with a square-type middle, and T_1 is odd for 209 991 and 117 477 414 815, so that these two kernels carry no triple at all (Lemma I.1.4).*

Proof. If $n = X^2$, then $X^4 - mU_k^2 = 1$. By Cohn's theorem [20], quoted as Theorem 1.1 of Bennett and Walsh [8], the only possible solutions are $X^2 = T_1$ and $X^2 = 2T_1^2 - 1 = T_2$. So $k \in \{1, 2\}$, and k is odd (Lemma I.1.1), so $k = 1$. If $n = 2X^2$, then $T_k = 2X^2$, and by Theorem 1.2 of Bennett and Walsh [8] at most one index k has this property, namely the smallest k with $2 \mid T_k$. As T_1 is even for a triple, this is $k = 1$. A fourth power is a square. In all cases $k = 1$ is a multiple of m' , so $m' = 1$ and $m \mid U_1$ by Lemma I.1.2. The squarefree m with $m \mid U_1(m)$ are listed in OEIS A135735, completely up to $1.5 \cdot 10^{12}$ [52, Remark 5.5] and, under the stated assumption, up to $5.325 \cdot 10^{13}$ [53] (Remark I.2.5); the four values above are the members with $m \equiv 7 \pmod{8}$, the first three below the smaller bound. For $m = 209\,991$ and for $m = 117\,477\,414\,815$ the number T_1 is odd (computed with exact integers), so these two kernels carry no triple at all by Lemma I.1.4. For the other two, a prime divides T_1 exactly once: 701 for $m = 4\,099\,215$ and 5 for $m = 39\,028\,039\,587\,479$. So $n = T_1$ is not powerful. $\square$

Cohn's theorem sharpens Ljunggren's bound of at most two solutions for $x^4 - Dy^2 = 1$ [36]. Cohn obtains the single case with two solutions, $D = 1785$, from Ljunggren's result that $y^2 = 2x^4 - 1$ has only the solutions $x = 1$ and $x = 13$ [37].

In words. If the middle of a triple is a square, twice a square, or a fourth power, then its kernel must divide U_1 . Such kernels are known completely up to $1.5 \cdot 10^{12}$, and up to $5.325 \cdot 10^{13}$ if a search that its author could not repeat missed nothing; none of the known ones carries a triple with such a middle. So a triple of this special shape would need an enormous kernel. The theorem does not exclude these middles; it restricts where they could be.

Remark I.5.5 (the share of the candidates covered). [computed] Theorem I.5.4 applies to a fixed share of the candidates for a middle. The middle of a triple is powerful and divisible by 4 (Lemma I.1.4), so we compare with all powerful n with $4 \mid n$. The number of powerful $n \leq x$ is asymptotic to $(\zeta(3/2)/\zeta(3))\sqrt{x}$ [24, 6]. The condition $4 \mid n$ multiplies this by $(\sqrt{2} + 1)/(2\sqrt{2} + 1)$: in the Dirichlet series of the powerful numbers, the Euler factor at 2 changes from $1 + 2^{-2s}/(1 - 2^{-s})$ to $2^{-2s}/(1 - 2^{-s})$, and at $s = 1/2$ their quotient is this number. Among these n , the squares number about $\sqrt{x}/2$ and the twice squares about $\sqrt{x}/(2\sqrt{2})$. Hence, as $x \rightarrow \infty$, the share of squares tends to

$$\frac{(2\sqrt{2} + 1)\zeta(3)}{2(\sqrt{2} + 1)\zeta(3/2)} = 0.3648,$$

the share of twice squares to 0.2580, and together to 0.6228. A count up to 10^{14} gives 0.6252. Cubes and higher odd powers have share tending to 0.

In words. Among all powerful numbers divisible by 4, which are the candidates for a middle, a little more than three fifths have the special shape of Theorem I.5.4. For these, a triple would need a kernel beyond $1.5 \cdot 10^{12}$, and beyond $5.325 \cdot 10^{13}$ under the assumption of Remark I.2.5.

6 Data

Proposition I.6.1 (exponents at the rank). [measured] *Consider the 750 kernels $m \equiv 7 \pmod{8}$ with $m \leq 10^4$ and $T_1(m)$ even, and the primes $3 \leq p \leq 3000$ that have a rank $\alpha_m(p)$; this gives 120 373 pairs (m, p) . Among them $v_p(T_{\alpha_m(p)}) \geq 2$ holds 470 times and $v_p(T_{\alpha_m(p)}) \geq 3$ holds 52 times. The model in which every pair has $v_p \geq 2$ with probability $1/p$ and $v_p \geq 3$ with probability $1/p^2$, independently, predicts 486 and 49.1.*

The model is the usual heuristic for Wieferich primes: Crandall, Dilcher and Pomerance record that the probability that p is a Wieferich prime to base 2 is sometimes estimated as about $1/p$ [21], and Ryan [57, Section 1] states it in the same form for the Wieferich locus of an arbitrary algebraic unit, conditional on an equidistribution of Fermat quotients that is open even for base 2. In our setting the unit is ε_m and the event is $p^2 \mid T_{\alpha_m(p)}(m)$, the Wieferich condition of Proposition I.6.10.

Both counts lie within one standard deviation of the prediction. The independence in the model was tested rather than assumed: all events of one kernel come from the same unit ε_m , so they might cluster. Let k_m be the number of events $v_p \geq 2$ at the kernel m and λ_m the sum of $1/p$ over its pairs. The statistic $S = \sum_m (k_m - \lambda_m)^2$ is 411.7, against 436.2 ± 28.2 in 4000 simulations of the independent model: $z = -0.87$, one-sided empirical $p = 0.80$. No clustering within kernels is detected. The kernels with 0, 1, 2, 3, 4 events number 396, 255, 83, 15, 1. Split by prime, the largest deviation is $p = 181$ at 2.4 standard deviations, about what testing each prime with an expected count of at least 1 separately produces by chance; primes with a smaller expected count were not ranked.

The power of the test was measured as well, with data that keep the frequencies of the events but give a common shock to a fraction ρ of the kernels (40 runs for each ρ). Full coupling, $\rho = 1$, lifts the mean z to about 652. At $\rho = 0.2$ the coupling is detected in 90 % of the runs, at $\rho = 0.05$ in 48 %. So a coupling of strength 0.2 or more would very likely have been seen, and a coupling of strength 0.05 or less cannot be decided from these data.

In words. For a prime p the chance that p^2 divides the first term of the Pell sequence divisible by p should be about $1/p$. Over the 120 373 pairs (m, p) this simple rule predicts the observed number of such squares and cubes well. We also checked that the squares do not come in bunches at single kernels, and we measured how strong a bunching the check would have seen.

Proposition I.6.2 (no triple below $1.83 \cdot 10^{18}$, and the pairs at distance two with middle divisible by four). [computed]

- (a) *The middles $n \equiv 0 \pmod{4}$ of pairs $n - 1, n + 1$ of powerful numbers with $n < 10^{2000}$ and $\text{sqfree}(n^2 - 1) \leq 10^8$ are exactly the numbers $T_k(m)$ of the 371 lattice points of Theorem I.3.1 with T_k even.*

- (b) Alekseyev's table for OEIS A076445 [47] has 33 terms up to its largest, which has 67 digits. Up to that term they are exactly the pairs at distance two of our computation with kernel at most 10^8 , for middles $\equiv 0$ and $\equiv 2 \pmod{4}$ alike. Exactly 9 of them have a middle $\equiv 0 \pmod{4}$.
- (c) The list in (a) contains every middle $n \equiv 0 \pmod{4}$ of a pair at distance two with $n < 1.83 \cdot 10^{18}$, and every such middle with $n < 3.88 \cdot 10^{20}$ if Reinhart's extended search missed nothing (Remark I.2.5). It contains every such middle with $n < 10^{36}/27$ whose kernel does not divide U_1 , and every such middle with $n < 10^{36}/27$ if no squarefree $m \equiv 7 \pmod{8}$ with $m \mid U_1(m)$ and $1.5 \cdot 10^{12} < m < 10^{24}/9$ has $T_1(m) < 10^{36}/27$.
- (d) Consequently, with Theorem I.3.1, no triple of consecutive powerful numbers has a middle below $1.83 \cdot 10^{18}$, and none below $3.88 \cdot 10^{20}$ if Reinhart's extended search missed nothing.

Proof. (a) is Lemma I.1.3 together with the enumeration of Theorem I.3.1: a lattice point with T_k odd gives no middle divisible by four. (b) is a comparison of the two lists; we compared our own values with the table through their hashes, so no value of the table enters our data. (c) Let n be such a middle and m its kernel. If $m \nmid U_1$, Corollary I.2.4(i) gives $m < (27n)^{2/9}$, which is at most 10^8 for $n < 10^{36}/27$. If $m \mid U_1$, part (ii) gives $m < n^{2/3}$ and, since $n = T_k(m) \geq T_1(m)$, also $T_1(m) \leq n$. For $n < 1.83 \cdot 10^{18}$ this forces $m < 1.5 \cdot 10^{12}$, the range in which Reinhart's list of such kernels is complete [52, Remark 5.5]; its 3 kernels $m \equiv 7 \pmod{8}$ all have $T_1(m) \geq 10^{94}$, so none of them occurs. For $n < 3.88 \cdot 10^{20}$ the same holds with the 4 kernels of the extended list [53], if that list is complete (Remark I.2.5). For $n < 10^{36}/27$ the same argument leaves exactly the kernels named in the condition. (d) The middle of a triple is such a middle (Lemma I.1.1), so by (c) it is in the list of (a), and every point of that list is excluded in Theorem I.3.1. $\square$

In words. Our list of the middles divisible by four is complete in the box of Theorem I.3.1, and it agrees with the longest published table in both directions. How far it is complete beyond the box depends on the second kind of kernel, the ones dividing U_1 : they are known with proof only up to $1.5 \cdot 10^{12}$, which gives completeness up to $1.83 \cdot 10^{18}$ without any assumption, and up to $3.88 \cdot 10^{20}$ if the extended search missed nothing.

Bloom [10] states that no triple has a middle below $7.38 \cdot 10^{28}$, by OEIS A076445. This rests on the terms listed there, and the entry does not state that its list is complete; Mian and Siddique make the same point [43].

Proposition I.6.3 (ladders for the known kernels dividing U_1). [computed] *Table 3 gives, for the two known squarefree kernels $m \equiv 7 \pmod{8}$ with $m \mid U_1$ and $T_1(m)$ even, the ladder of Theorem I.5.1: at each stage the primes $p \leq P$ with $v_p(T_k) = 1$ at the current index k , and then the smallest index a triple with this kernel could still have, together with the number of digits its middle would have at least. The other two known kernels, $m = 209\,991$ and $m = 117\,477\,414\,815$, have T_1 odd (with 95 and 11 621 digits) and give no triple by parity.*

How the table is computed. Let k_0 be an index with $v_p(T_{k_0}) = 1$ for an odd prime p , and let T_k be powerful with $k_0 \mid k$ and k/k_0 odd. By Lemma I.4.3, $v_p(T_k) = 1 + v_p(k/k_0) \geq 2$, so $p \mid k/k_0$. Starting at $k_0 = 1$, the product of the witnesses of a stage therefore divides k/k_0 ,

kernel m	digits of T_1	P	witnesses, stage by stage	$k \geq$	digits of $n \geq$
4 099 215	228	$2 \cdot 10^5$	701, 100469, 157049	$1.1 \cdot 10^{13}$	$2.5 \cdot 10^{15}$
39 028 039 587 479	1 880 030	$3 \cdot 10^3$	5, 7, 71, 1657; 19, 29, 421, 569, 1987; 379, 2089, 2203, 2437	$4.5 \cdot 10^{30}$	$8.6 \cdot 10^{36}$

Table 3: The ladder for the two known kernels $m \equiv 7 \pmod{8}$ with $m \mid U_1$ and $T_1(m)$ even. The witnesses of one stage are separated by commas, the stages by semicolons. The last two columns are rounded down.

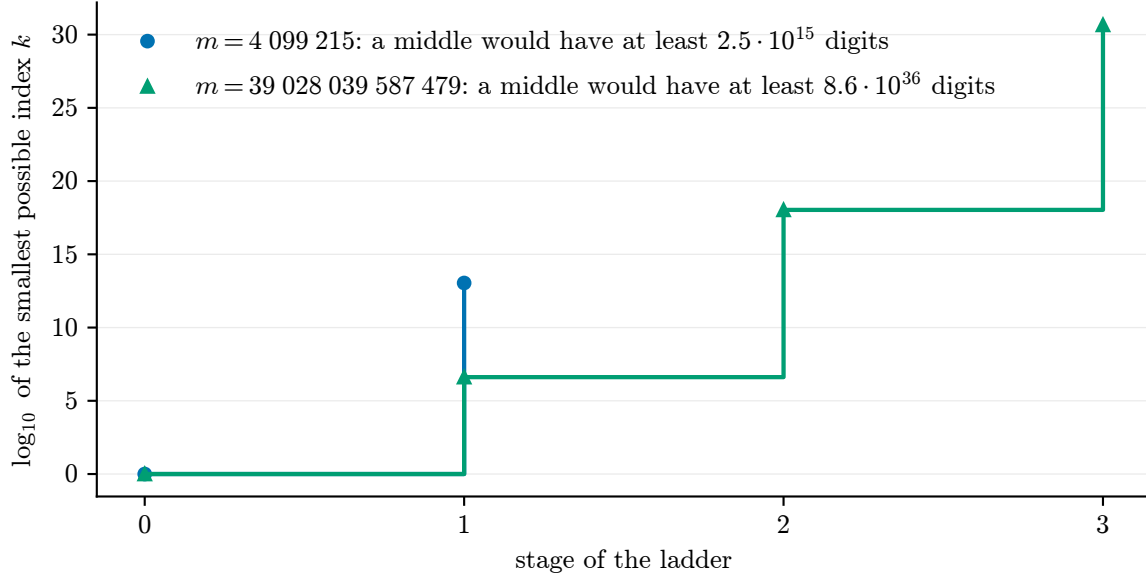


Figure 4: **Each stage of a ladder pushes the smallest possible index up.** The ladder of Table 3, stage by stage. Each step multiplies the smallest possible index k by the product of the witnesses of that stage; the last point is the smallest index that remains possible.

and the next stage starts at k_0 times that product. The witnesses are found from T_1 and U_1 modulo p^2 , without large integers. The ladder stops at the first stage without a witness below P . $\square$

In words. These two kernels are dangerous in the sense of Section 2, because they divide U_1 ; the other two known kernels divide U_1 as well, but their T_1 is odd. For each of them the table shows how far out a triple would have to be: the middle would have at least the number of digits in the last column. For $m = 4\,099\,215$ the indices $k = 1, 3, 5, 7$ also lie in the box of Theorem I.3.1, where they are excluded by the witnesses 701, 11, 19, 29.

Reinhart [53, Def. 1.4, p. 6] asks whether 4 099 215 or 39 028 039 587 479, or even 7, admits an index k with T_k powerful and $m \mid U_k$ (his condition (C), Question I.9.4). Our first witnesses 701 and 5 for these two kernels are his own. The table does not decide the question; it only says how large a positive answer would have to be.

Proposition I.6.4 (the smallest witnesses). [computed] *For each of the 371 lattice points of Theorem I.3.1 that are excluded by a witness, let p be the smallest witness and $\alpha_m(p)$ its*

rank. In 8 points the smallest witness is primitive, $\alpha_m(p) = k$; in all others it is a prime whose rank divides k with odd cofactor. The smallest witness is 29 in 79 points, 19 in 78, 11 in 45, 3 in 43, 5 in 33 and 7 in 25; its median is 19 and its maximum 114 997. In every point $v_p(T_{\alpha_m(p)}) = 1$ and $p \nmid k/\alpha_m(p)$.

The valuation identity of Lemma I.4.3 was checked in every point. As controls, the point (7, 7) has the witness 29 of rank 7, and the identity predicts $v_{197}(T_{1379}(7)) = 2$, which the exact computation confirms.

In words. A triple would have to survive the test of Theorem I.4.6 not only at the primes that first appear at its index k , but at every small prime whose rank divides k . In practice the lattice points fail at small primes: the typical witness is 19 or 29, and only at 8 of the points is the smallest witness a prime that first appears at k itself.

Remark I.6.5 (the condition modulo 36 and the witnesses). [computed] By Beckon's condition (Remark I.8.1) the middle n of a triple satisfies $n \equiv 0, 8$ or $28 \pmod{36}$; the earliest derivation we located is Beckon's [7, Theorem 0.1]. Proposition 3(i) of [59] refines it to 39 classes modulo 900 and 1209 classes modulo 44 100. Rizzi [55, Proposition 5.1] derives the condition modulo 36 again and notes that it agrees with Beckon's; the same paper reports a search up to 10^{10} that finds fourteen pairs and no triple [55, Chapters 4 and 6]. Of the 371 middles of Proposition I.6.4, 328 satisfy the condition, and the 43 that do not are exactly the points whose smallest witness is 3.

This is no coincidence. For the middle n of a pair, that is, with $n - 1$ and $n + 1$ powerful, and $4 \mid n$, the condition fails exactly when n is divisible by 3 but not by 9, that is, when $v_3(T_k) = 1$; then 3 is a witness, and the smallest one. So the condition modulo 36 is the case $p = 3$ of the witness reduction. Since the two computations work by different methods, their agreement on the same points checks both.

In words. A known rule says what the middle of a triple must look like modulo 36. Our witness search already contains this rule: the middles that break it are exactly those where the prime 3 appears only once.

Proposition I.6.6 (how many conditions there are, and how often they are met). [computed]

- (a) For the 15 kernels of class 7 with the most lattice points and all primes $p \leq 10^6$ that have a rank d , the congruence $p \equiv \pm 1 \pmod{4d}$ of Corollary I.4.8(i) holds at all 425 362 pairs (p, d) .
- (b) Over the 445 lattice points of Theorem I.3.1, the indices k have 2149 divisors $d \geq 5$ in total (median 4 per point, maximum 16). For 1682 of them (78.3 %) one has $k < d(4d - 1)$, so by Corollary I.4.8(iii) a powerful T_k would need $p_d^2 \mid T_d$ at that d . Per point the number of such conditions has median 3, mean 3.8 and maximum 12.
- (c) For the 25 kernels of class 7 with the most lattice points (378 points), every divisor of every k was taken as a rank, and for each rank the first 5 primes of that rank below $3 \cdot 10^6$ were tested. Of the 2536 rank occurrences (a point together with a divisor of its index, so that $d = 1$ and $d = 3$ are included and one pair (m, d) counts once for every point whose index d divides), 2025 have a prime below this bound, and 1988 of these (98.17 %) carry a witness; 37 carry only primes with exponent at least 2. Testing only

the smallest prime of each rank gives 1936 of the same 2025 ranks (95.60 %) and 89 with exponent at least 2 only.

The 511 rank occurrences without a prime below the bound are undecided; for $d \geq 5$ they are not known to be free of witnesses, since their smallest prime lies higher, and among them are the occurrences of $d = 1$ at kernels with $T_1(m)$ a power of 2, which have no odd prime at all. Parts (b) and (c) report rates. No proof in this paper depends on them, and Corollary I.4.8 holds independently of them; the single ranks behind (c) are decided, since one prime of exponent 1 settles its rank by Theorem I.4.6.

In words. Part (a) checks a proved rule on 425 362 cases. Part (b) counts how many separate tests a triple would have to pass at a lattice point: typically a few, sometimes a dozen. Part (c) shows how often such a test is actually decided by a small prime: almost always, and more often when more primes per rank are tried.

Proposition I.6.7 (middles that are powers of two). [computed] *For every integer a with $2 \leq a \leq 100\,000$, at least one of $2^a - 1$ and $2^a + 1$ is not powerful. In particular no triple has a middle 2^a in this range, that is, a middle of this form with up to 30 103 digits.*

The computation. For each prime $p \leq 10^6$ let e be the order of 2 modulo p . The prime p divides $2^a - 1$ exactly when $e \mid a$, and then $v_p(2^a - 1) = v_p(2^e - 1) + v_p(a/e)$; so p is a witness for $2^a - 1$ exactly when $e \mid a$, $p^2 \nmid 2^e - 1$ and $p \nmid a/e$. The prime p divides $2^a + 1$ exactly when e is even and a is an odd multiple of $e/2$, and then $v_p(2^a + 1) = v_p(2^{e/2} + 1) + v_p(2a/e)$; so p is a witness for $2^a + 1$ exactly when e is even, a is an odd multiple of $e/2$, $p^2 \nmid 2^{e/2} + 1$ and $p \nmid 2a/e$. The loop runs over the primes rather than over the exponents. Among the 99 999 exponents, 8957 (8.96 %) leave $2^a - 1$ without a witness below the bound and 2511 (2.51 %) leave $2^a + 1$ without one; no exponent leaves both without one. $\square$

The exponents without a witness are undecided; they are not known to be powerful: their prime factors lie above the search bound. Equivalently, $2^a - 1$ is proved not powerful for 91 042 of the exponents and $2^a + 1$ for 97 488. As controls, the computation of the orders returns exactly the two Wieferich primes 1093 and 3511 below the bound, and $2^3 + 1 = 9$, which is powerful, receives no witness. Question I.9.3 asks for all a .

In words. If the middle of a triple were a power of two, then $2^a - 1$ and $2^a + 1$ would both be powerful. For every a up to 100 000 a small prime appears exactly once in one of the two numbers. The search runs over primes, which makes it cheap.

Proposition I.6.8 (the primitive part over an explicit finite set). [computed] *Let $\mathcal{P}$ be the set of the 445 lattice points of Theorem I.3.1; all of them are excluded there. For a kernel m of a point of $\mathcal{P}$ and an odd $d \geq 5$ dividing an index k with $(m, k) \in \mathcal{P}$, let M_d be the part of $T_d(m)$ supported on the primes of rank exactly d . There are 961 such pairs (m, d) , on the 79 kernels of $\mathcal{P}$.*

(a) Settled: 954 of the 961, each by a certified witness.

(b) Open: 7 of the 961.

A witness here is a prime p of rank d with $v_p(M_d) = 1$; it shows that M_d is not powerful. For some blocks the witness comes from a complete factorization, and for one from the

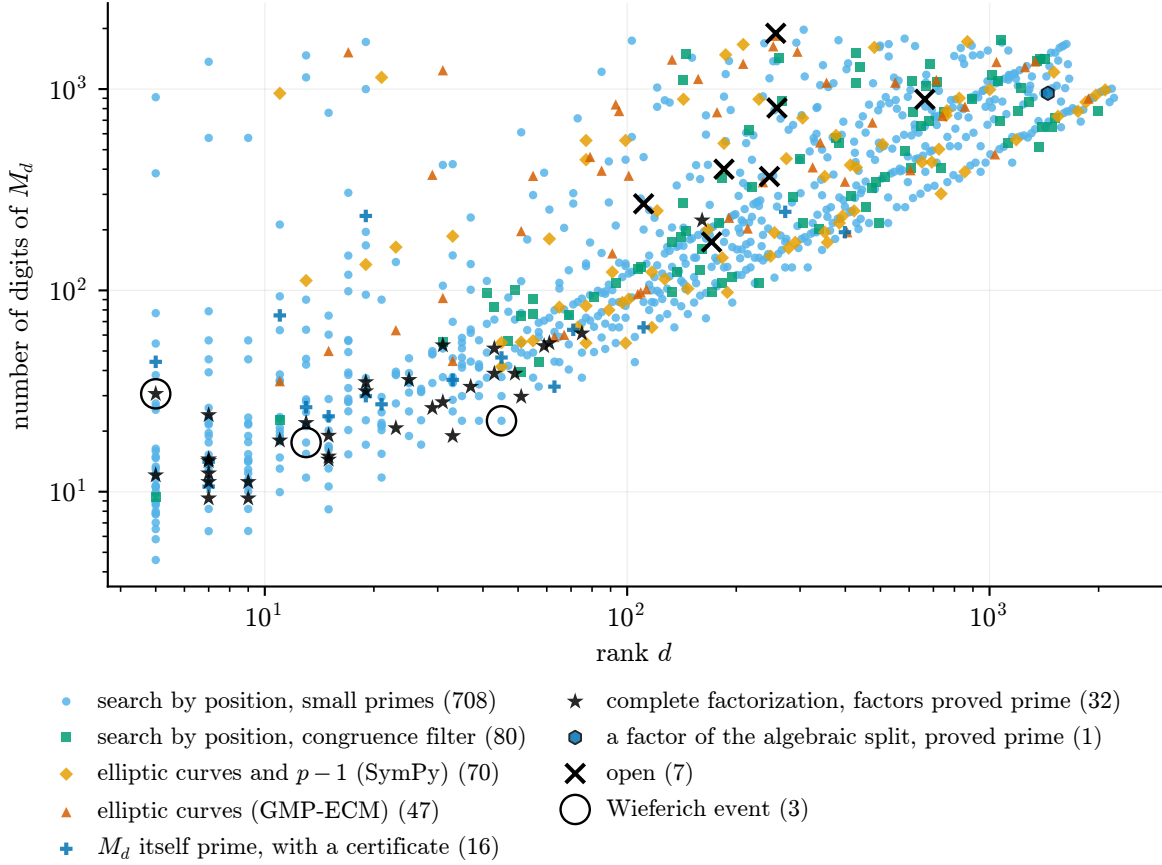


Figure 5: **Every building block, settled or open.** The 961 building blocks M_d of Proposition I.6.8, one point each: the rank d against the number of digits, both on logarithmic scales. Color and shape show how each block was shown not to be powerful; the classes are the rows of Table 4, and every class except the open blocks is a certified witness. The 7 open blocks (crosses) have at least 173 digits each, and no search found a factor of them. Circles mark the 3 blocks with a Wieferich event below $3 \cdot 10^6$ (Proposition I.6.10). The terms are explained in Table 2.

algebraic split of Remark I.6.13. For a prime p of rank exactly d one has $v_p(M_d) = v_p(T_d)$ by definition; for the Möbius dual $\prod_{e|d} T_e^{\mu(d/e)}$, the product with which the computations work, the same holds at every prime of rank exactly d [56, Theorem 2.2(c)]; the restriction to odd d is the range in which the dual of the companion sequence is integral [56, Theorem 2.3]. Since every point of $\mathcal{P}$ is already excluded, the proposition excludes no further triple. It asks, on these pairs, whether a primitive part can be powerful (Question I.9.2). For 21 of the kernels $T_1(m)$ is odd; their points are excluded by parity alone and enter only as instances of that question.

How the pairs were settled. A search by position. Every prime $p \leq 3 \cdot 10^6$ of rank d was

how the pair was settled	first 25 kernels	other 54	all
certified witness: search by position, small primes	605	103	708
certified witness: search by position, congruence filter	64	16	80
certified witness: elliptic curves and $p - 1$ (SymPy)	70	0	70
certified witness: elliptic curves (GMP-ECM)	27	20	47
certified witness: M_d itself prime, with a certificate	8	8	16
certified witness: complete factorization, factors proved prime	32	0	32
certified witness: a factor of the algebraic split, proved prime	1	0	1
open	4	3	7

Table 4: How the 961 pairs of Proposition I.6.8 were settled. The first 25 kernels are those with the most lattice points; for the other 54, and for the pairs of the first 25 with $t_{35} < 1$, the elliptic curve method ran to a smaller effort, see Table 5.

tested, and later, restricted by Corollary I.4.8(i) to $p \equiv \pm 1 \pmod{4d}$, every such prime up to 10^9 . For the pairs that remain open, this search also shows that no prime of rank d with $v_p(M_d) = 1$ lies below its bound. *A search by size.* The elliptic curve method, first with SymPy [42] and then with GMP-ECM [76], looks for a factor by its size rather than by its position; it shows only what it finds. The effort spent on the pairs that remain open is in Table 5. *Complete factorization.* Small primitive parts were factored completely, and some are themselves prime, which no search for a proper divisor can see. One pair, $(m, d) = (319, 5)$, has no witness below the first bound and has its own row in Table 4; its witnesses come from the complete factorization of $M_5(319)$ given after Proposition I.6.10, whose large factors are proved prime. *Checks.* Every witness was recomputed on a second, independent route (the powers of ε_m with the rank taken from the order of the unit, or the Lucas sequence $V_n(2T_1, 1)$, whichever differs from the route that found it). Every witness of the 954 pairs in the first part of (a) is certified: below ψ_{12} by the Miller–Rabin test with the first twelve prime bases, which is deterministic there [65], and above it by Pocklington’s criterion from a factorization of $p - 1$ [12, Theorem 4 and Corollary 1], each certificate checked by a separate program. The prime factors of the complete factorizations and the factor from the algebraic split are certified in the same way: below ψ_{12} by the deterministic test, and above it 18 of them, with up to 75 digits, by Pocklington’s criterion, and 6, with up to 477 digits, by elliptic curve certificates [27, 2], which PARI/GP [70] produced and a separate program of ours checked step by step. $\square$

In words. Theorem I.3.1 already rules out all 445 lattice points in its search range. This proposition asks a different question: can a building block M_d be powerful at all? We take all 79 kernels of the lattice points and look at every building block that occurs there, 961 in total. A number is powerful when each of its prime factors appears at least twice. So one prime that appears exactly once is enough to show that a block is not powerful. We found such a prime, and proved that it is prime, for 954 blocks; for some of them we split the whole block into its prime factors, and for one the algebraic split of Remark I.6.13 gives the factor. Every prime used is proved prime, the large ones by certificates that our own program checks. For the last 7 blocks we found no such prime. Finding nothing only means that our search did not succeed, not that no such prime exists. The 7 open blocks are in the deposit with their decimal expansions; a prime factor of any one of them with exponent one settles that

m	d	digits	curves at $B_1 =$			t_{35}	t_{40}
			$5 \cdot 10^4$	10^6	$3 \cdot 10^6$		
399	171	173	300	1071	2753	8.36	1.10
407	111	269	300	1071	2753	8.36	1.10
39	247	367	300	1071	2753	8.36	1.10
111	185	400	300	1071	2753	8.36	1.10
407	259	805	300	1071	2753	8.05	1.05
663	663	889	300	1071	38	1.05	0.11
20 303	257	1893	300	209	–	0.22	0.02

Table 5: The 7 open pairs of Proposition I.6.8 and the effort of GMP-ECM on each. The columns t_{35} and t_{40} give the fraction of the curves the program expects to need for a factor of 35 and of 40 digits, for the second-stage bound it chose. A fraction of 1 still misses a factor of that size with probability about $1/e$, so “none found” describes the effort, not the factors. The kernels 399, 663 and 20 303 are among the other 54; the rest are among the first 25. The blocks that the searches left open on the kernels $m = 7$ and $m = 23$, where $T_1(m) + 1$ is a square, are settled by the algebraic split of Remark I.6.13 and are not listed.

block, and a complete factorization settles it either way.

A different question, measured. Among the 1341 triples (m, d, p) of the search by position with $p \leq 3 \cdot 10^6$ (for each rank, the first five primes found by the search), $v_p = 1$ in 1338 cases, $v_p = 2$ in 3 and $v_p \geq 3$ in none. The heuristic $P(v_p \geq j) = p^{1-j}$ predicts 2.38 triples with $v_p \geq 2$. The observations are consistent with it and do not confirm it.

Remark I.6.9 (where this stands in the literature). [literature] Ribenboim and Walsh [54, Corollary 2] show that the abc conjecture implies that a Lucas sequence of positive discriminant has only finitely many powerful terms, and Yabuta [74] extends this to negative discriminant. These statements concern the terms rather than their primitive parts, and they are conditional and not effective, so they neither contain nor are contained in the computation above. Closer to the question is [56, Corollary 5.1]: the primitive parts of the Fibonacci numbers are squarefree for all $n \neq 6$ if and only if there is no Wall–Sun–Sun prime. Square-free is a far stronger property than not powerful; the question asked here is the weaker one, and it is the one a finite computation can address.

In words. If the abc conjecture is true, only finitely many terms of such sequences are powerful, but nobody knows which, and the statement is about whole terms, not their building blocks. For the Fibonacci numbers, even the question whether all building blocks are squarefree is tied to an open problem about special primes.

Proposition I.6.10 (Wieferich primes of the unit and the primitive parts). [proved] *Fix a squarefree $m \equiv 7 \pmod{8}$ with $T_1(m)$ even. For odd $d \geq 5$ let M_d be the part of $T_d(m)$ supported on the primes of rank exactly d . Call an odd prime $p \nmid m$ that divides some $T_k(m)$ a Wieferich prime for ε_m if $p^2 \mid T_{\alpha_m(p)}(m)$. The following are equivalent:*

- (i) ε_m has only finitely many Wieferich primes of odd rank;
- (ii) M_d is squarefree for all but finitely many odd $d \geq 5$.

In that case M_d is not powerful for all but finitely many odd $d \geq 5$. Moreover, let $p \nmid m$ be an odd prime that has a rank α , let $u_n = U_n/U_1$ be the Lucas sequence of the first kind for

$(P, Q) = (2T_1, 1)$, and let $e = (m/p)$. Then $p \nmid U_1$ and $v_p(u_{p-e}) = v_p(T_\alpha)$. So the Wieferich primes for ε_m are exactly the odd Lucas–Wieferich primes of this sequence that divide some T_k .

Proof. A prime p dividing M_d has rank exactly d , so $v_p(M_d) = v_p(T_d) = v_p(T_{\alpha_m(p)})$; in the language of Möbius duals this is [56, Theorem 2.2(c)]. Hence p divides M_d exactly once when p is not a Wieferich prime for ε_m . (i) $\Rightarrow$ (ii): a Wieferich prime of odd rank d affects only M_d , so under (i) all M_d outside finitely many odd d are squarefree. (ii) $\Rightarrow$ (i): a Wieferich prime of odd rank $d \geq 5$ makes M_d non-squarefree, so under (ii) such primes have one of finitely many ranks, and each rank carries finitely many primes, since M_d is a positive integer; the ranks 1 and 3 carry only the prime divisors of T_1 and T_3 . For the last assertion, $M_d > 1$ for odd $d \geq 5$, because T_d has a primitive prime divisor [15], in the form of [73] used in the proof of Corollary I.4.8; and a squarefree integer above 1 is not powerful. For the last statement, $p \mid T_\alpha$ gives $\varepsilon_m^{2\alpha} + 1 = 2\varepsilon_m^\alpha T_\alpha \equiv 0 \pmod{p}$ in $\mathbb{Z}[\sqrt{m}]$, and the proof of Corollary I.4.8(i) shows that ε_m has order 4α modulo the primes above p and that $4\alpha \mid p - e$; in particular $p \nmid U_1$, because $p \mid U_1$ would give $\varepsilon_m^2 \equiv 1$. Write $y = \varepsilon_m^{4\alpha}$ and $M = (p - e)/(2\alpha)$, so that $\varepsilon_m^{2(p-e)} = y^M$ and $y \equiv 1 \pmod{p}$. Here $p \nmid M$, because $p \nmid p - e$, and $y^M - 1 = (y - 1)(1 + y + \cdots + y^{M-1})$ with the second factor congruent to M modulo p . Multiplication by an element of $\mathbb{Z}[\sqrt{m}]$ whose norm is prime to p does not change v_p (see the proof of Lemma I.4.3), so $v_p(y^M - 1) = v_p(y - 1)$. Next, $y - 1 = (\varepsilon_m^{2\alpha} + 1)(\varepsilon_m^{2\alpha} - 1)$ with $\varepsilon_m^{2\alpha} - 1 \equiv -2 \pmod{p}$ of norm prime to p , so $v_p(y - 1) = v_p(2\varepsilon_m^\alpha T_\alpha) = v_p(T_\alpha)$. Finally $\varepsilon_m^{p-e} \cdot u_{p-e} \cdot 2U_1\sqrt{m} = \varepsilon_m^{2(p-e)} - 1$, and the norms of ε_m^{p-e} and $2U_1\sqrt{m}$ are prime to p , so $v_p(u_{p-e}) = v_p(\varepsilon_m^{2(p-e)} - 1) = v_p(T_\alpha)$. $\square$

In words. A prime p is a Wieferich prime (to base 2) if p^2 divides $2^{p-1} - 1$. Below 10^6 there are exactly two, 1093 and 3511 (the control in Proposition I.6.7). Fix one m with $T_1(m)$ even. The rank of a prime p is the first k with p dividing $T_k(m)$. We call p a Wieferich prime for m if p^2 already divides the term at its rank. If this rank d is odd and at least 5, then p^2 divides the block M_d , so M_d is not squarefree. Proposition I.6.10 says: the blocks M_d with odd d are squarefree from some point on exactly when only finitely many such primes have odd rank. In that case only finitely many blocks with odd d can be powerful. Such primes do exist. Among the primes below $3 \cdot 10^6$ we found 15 of odd rank $d \geq 5$ on those of the 79 kernels of Proposition I.6.8 that have T_1 even. Since finitely many are allowed, these do not settle the condition. For 12 of them the block is not powerful: another prime divides it exactly once. For the other 3 we do not know. Whether a block can be powerful remains open in general. For the Fibonacci numbers, the same kind of condition is the open question whether a Wall–Sun–Sun prime exists.

For the Fibonacci numbers, Ross, Shen and Cai show that the Möbius duals $\prod_{d|n} F_d^{\mu(n/d)}$ are squarefree for all $n \neq 6$ exactly when there is no Wall–Sun–Sun prime [56, Corollary 5.1]. The proposition is the analog for ε_m , restricted to odd rank and allowing finitely many exceptions; it is elementary, and its content is the identification of the hypothesis.

Remark I.6.11 (the Wieferich events of the units). [computed] That hypothesis is not empty for our units. Among the primes below $3 \cdot 10^6$ the 79 units of Proposition I.6.8 carry 54 Wieferich events, that is, pairs of a unit and a prime (40 distinct primes). Of these, 18 have odd rank $d \geq 5$, and 15 of those lie on units with T_1 even, where the proposition applies;

Table 6 lists them. So the squarefree statement for all odd d fails for these units. In 11 of the 15 cases M_d is nevertheless not powerful, since a prime of rank d below $3 \cdot 10^6$ divides it exactly once, and $M_5(319) = 20161^2 \cdot 1\,375\,502\,561 \cdot 158\,585\,313\,121$ with both large factors proved prime. For the remaining 3 the question is not settled here; they lie outside the finite set of Proposition I.6.8. Conversely, the 3 events whose block lies in that set, the circles in Figure 5, are all settled; the last column of Table 6 marks them. On the 25 units of the first search there are 16 events. Part III finds more on the same units (Remark III.4.1 in [Part III]) because it also counts primes that divide no $T_k(m)$. Whether a primitive part can be powerful at all is Question I.9.2.

unit m	rank d	prime p	M_d not powerful by	in Prop. I.6.8
15	45	181	witness	yes
31	39	157	witness	no
79	253	4049	witness	no
143	13	311	witness	yes
223	45	179	witness	no
319	5	20 161	complete factorization	yes
383	21545	258 539	witness	no
447	413	24 781	witness	no
455	9	37	witness	no
615	807	3229	not settled	no
1343	5	19	witness	no
1535	63	251	witness	no
2135	13	53	witness	no
2135	37	149	not settled	no
123 879	9	37	not settled	no

Table 6: The Wieferich events of odd rank $d \geq 5$ below $3 \cdot 10^6$ on the units ε_m of Proposition I.6.8 with $T_1(m)$ even. “Witness”: a prime of rank d below $3 \cdot 10^6$ divides M_d exactly once. “Not settled”: not decided by our computations. The last column says whether (m, d) is one of the 961 pairs of Proposition I.6.8.

Remark I.6.12 (Lucas–Wieferich primes). [computed] The condition in (i) has a name in the computational literature. For the Lucas sequence (u_n) of the first kind attached to $(P, Q) = (2T_1, 1)$, a prime p with $u_{p-e} \equiv 0 \pmod{p^2}$, where e is the Legendre symbol of the discriminant, is a Lucas–Wieferich prime [41, p. 2088]. That condition is placed on the sequence of the first kind and ours on the companion sequence, but for primes that have a rank the two agree, by the last statement of Proposition I.6.10. The computation agrees as well: on the 16 events of the first search of Proposition I.6.8 (the 25 units with the most lattice points) the two conditions select the same primes; as controls, the 3 Wieferich primes 13, 31, 1546463 of the Pell sequence are recognized, and 25 primes that divide their term exactly once are not. This checks the computation on one sequence; the identity itself is proved above.

Remark I.6.13 (a second primitive prime factor). [computed] If every block M_d with $d \geq 5$ had two distinct prime factors, a powerful block would need two primes of rank d that each

divide $T_d(m)$ at least twice, two simultaneous events of the kind counted in Proposition I.6.1. For Lehmer numbers there is a theorem of this kind: Juričević [31, Theorems 1.1 and 2.1], making a result of Schinzel effective, proves that the n -th term of a real Lehmer pair has at least two primitive prime divisors when $n > 4$, $n \neq 6$ and $n/(\eta\kappa)$ is odd, where η and κ are invariants of the pair, apart from an explicit list of exceptions with $n \leq 30$. Here M_d is the primitive part of the term of index $2d$ of the Lucas sequence $u_n(2T_1, 1) = U_n/U_1$, whose terms of even index are $2T_1$ times the Lehmer numbers of the pair with $L = 4T_1^2$ and $M = 1$. For this pair $\eta = \kappa = 1$, so the condition asks for $2d$ to be odd, and the theorem does not apply to any block.

A second pair. If $T_1(m) + 1 = x^2$ is a square, then $\varepsilon = \gamma^2$ with $\gamma + \gamma^{-1} = x\sqrt{2}$, and (γ, γ^{-1}) is a real Lehmer pair with $L = 2x^2$ and $M = 1$, for which $\kappa = 2$ and $\eta = 2$. Its term of index $4d$ is the term of index $2d$ above, so its primitive divisors are the primes of rank d , and for odd $d \geq 5$ the quotient $4d/(\eta\kappa) = d$ is odd; none of the exceptional triples has these values. Hence on such a kernel every block M_d with $d \geq 5$ has at least two distinct prime factors. The split behind it is elementary: for odd d , $T_d + 1 = (xw_d)^2$ with $w_d = (\gamma^d + \gamma^{-d})/(\gamma + \gamma^{-1})$ an integer, so $T_d = (s - 1)(s + 1)$ with $s = xw_d$, and the odd number M_d is the product of the coprime factors $\gcd(M_d, s - 1)$ and $\gcd(M_d, s + 1)$; that M_d has two distinct prime factors is the content of the theorem. This is a splitting of the same kind as the Aurifeuillian factorization of the Lucas numbers, where the primitive part of L_{5n} , for odd n , is the product of two coprime factors [13, (2.9)–(2.11)]. The case occurs for 26 of the 79 kernels of Proposition I.6.8, which carry 361 of the 961 blocks, and for every prime kernel $m \equiv 7 \pmod{8}$: the class number is odd, so $x^2 - my^2 = 2$ is solvable (Remark I.4.11 gives an elementary proof), and we checked it for the 171 prime kernels below 5000. We also checked the conclusion on each of these 361 blocks: M_d is composite and not a prime power.

On the other kernels the theorem does not apply, and the data show that a block can have a single prime factor: of 163 blocks M_d with $d \geq 5$ and small $T_d(m)$ that we factored completely, 18 are prime, none of them on a kernel of the kind above, the largest at $d = 59$, and 5 of these have $2d \geq 31$, so that it is the parity condition and not the size of n that fails. So on the 26 kernels a powerful block needs two rare events, and on the others no argument of this kind forces two events: a block can have a single prime factor, and a powerful block could then be the power of a single prime.

The split also settles blocks that the searches of Proposition I.6.8 left open. For $(m, d) = (23, 161)$ both factors $\gcd(M_d, s - 1)$ and $\gcd(M_d, s + 1)$, with 111 and 112 digits, are prime, so M_d is their product and is not powerful. For $(m, d) = (7, 1449)$ one factor, with 477 digits, is prime; since the two factors are coprime, it divides M_d exactly once. All three primes are proved by elliptic curve certificates (proof of Proposition I.6.8).

In words. One could hope that a building block always has at least two different primes, so that a powerful block would need two rare events at once. A theorem of this type exists for a related sequence. On the kernels for which $T_1 + 1$ is a square, it applies after a change of the sequence, and there every block has at least two primes. On the other kernels its condition is never met by our blocks, and in our data some blocks are themselves prime. So there one rare event may be all that a powerful block needs.

7 Profile of a counterexample

This section collects in one place what is known about a triple, if one exists. Every clause is proved above or computed in Section 5 or 6, and the clauses that depend on Reinhart's extended search say so; nothing new is claimed here except clause (viii).

Theorem I.7.1 (profile). [computed] *Suppose that $n - 1$, n and $n + 1$ are all powerful, let $m = \text{sqfree}(n^2 - 1)$, and let (m, k) be the address of n .*

- (i) $m \equiv 7 \pmod{8}$ is squarefree, $T_1(m)$ is even, k is an odd multiple of m' , $n = T_k(m)$, and $4 \mid n$.
- (ii) $n \geq (m^{3/2}/m')^{m'}$. If $m \nmid U_1$, then $n > m^{9/2}/27$ and $k \geq 3$; if $m \mid U_1$, then $n \geq T_1 > m^{3/2}$.
- (iii) $v_2(n) = v_2(T_1(m))$, and k is divisible by every odd prime p with $v_p(T_1(m)) = 1$.
- (iv) Every odd prime $p \mid n$ is coprime to m , k is an odd multiple of $\alpha_m(p)$, and $v_p(T_{\alpha_m(p)}) + v_p(k/\alpha_m(p)) \geq 2$. Every prime p with $\alpha_m(p) = k$ satisfies $p^2 \mid n$.
- (v) Let $b = \text{sqfree}(n)$. If $b > 1$, then $k = \alpha_m(b)$, and $v_p(T_{\alpha_m(p)}) + v_p(\alpha_m(b)/\alpha_m(p)) \geq 3$ for every odd prime $p \mid b$.
- (vi) If n is a square, twice a square, or a fourth power, then $k = 1$, $m \mid U_1$, and $m > 1.5 \cdot 10^{12}$ or m is one of 209 991, 4 099 215 and 117 477 414 815, which carry no triple with such a middle; if Reinhart's extended search missed nothing (Remark I.2.5), then $m > 5.325 \cdot 10^{13}$ or m is one of the four known values.
- (vii) *Explicit exclusions.* Either $m > 10^8$ or $n \geq 10^{2000}$. If $m \leq 10^6$, then $\text{sqfree}(n) > 10^4$. If $m \leq 2000$, then n has more than $8.4 \cdot 10^7$ digits. If $10^8 < m \leq 10^{15}$ and m is a product of at least 4 primes up to 100, then $n \geq 10^{2000}$. If $m \mid U_1$ and $m \leq 1.5 \cdot 10^{12}$, then $m = 4\,099\,215$, which forces $k \geq 1.1 \cdot 10^{13}$ (the other members of Reinhart's list below this bound, 209 991 and 117 477 414 815, have T_1 odd and carry no triple); if Reinhart's extended search missed nothing, the same holds for $m \leq 5.325 \cdot 10^{13}$ with the further kernel 39 028 039 587 479 and $k \geq 4.5 \cdot 10^{30}$. In either case n has at least $2.5 \cdot 10^{15}$ digits. In all cases $n > 1.83 \cdot 10^{18}$ (Proposition I.6.2(d)).
- (viii) If $T_1(m) + 1$ is a square, then $n + 1$ is a square. This happens for 26 of the 79 kernels of Proposition I.6.8 and for every prime $m \equiv 7 \pmod{8}$ (Remark I.6.13).

Proof. (i) Lemmas I.1.1, I.1.3 and I.1.4, and Theorem I.4.6 for $4 \mid n$. (ii) Remark I.2.2 and Corollary I.2.3; if $m \nmid U_1$ then $m' \geq 3$, so $k \geq 3$. (iii) Lemma I.4.7. (iv) Theorem I.4.6 and Lemma I.4.3. (v) Lemma I.4.2 and Proposition I.4.4. (vi) Theorem I.5.4. (vii) Theorems I.3.1, I.4.5, I.5.1 and I.5.3; the kernels 209 991 and 117 477 414 815 have T_1 odd (Theorem I.5.4 and Proposition I.6.3), and the bounds for the other two come from the ladder of Theorem I.5.1 applied to these kernels, as reported in Proposition I.6.3. (viii) Remark I.6.13: $T_k + 1 = (xw_k)^2$ for odd k if $T_1 + 1 = x^2$. $\square$

In words. The theorem lists the conditions on a counterexample that this part proves or computes: clauses (i) to (v) are proved, (vi) and (vii) rest on computations. Its kernel m and its index k must pass every test proved in this paper at the same time: the parity, the height bounds, the exponent conditions, and the explicit searches.

The kernels with $m' = 1$, for which every odd k is admissible, are exactly those with $m \mid U_1$ (Lemma I.1.2); they are listed in OEIS A135735, completely up to $1.5 \cdot 10^{12}$ [52, Remark 5.5] and, if Reinhart's extended search missed nothing, up to $5.325 \cdot 10^{13}$ (Remark I.2.5). They are the weakest point for an unconditional lower bound on n : an unknown kernel of this kind above $1.5 \cdot 10^{12}$ gives only $n > m^{3/2} > 1.83 \cdot 10^{18}$ (the discussion after Corollary I.2.4 records that no stronger bound is available without hypotheses, and Remark I.4.12 discusses the *abc* route). The enumeration behind the b-file of OEIS A060355 is described by Mian and Siddique as exhaustive below 10^{22} but not certified [48, 43]. The terms of OEIS A076445 recorded by Bloom reach $7.38 \cdot 10^{28}$ [10, 47], and Alekseyev's table, which we compare in Proposition I.6.2(b), goes up to a term with 67 digits; neither list is known to be complete.

8 Routes that do not close the problem

Several natural approaches do not decide the problem. This section records why, so that the reasons are available together.

Remark I.8.1 (congruences). [literature] Beckon [7, Theorem 0.1] showed that the smallest member n of a triple $(n, n+1, n+2)$ of powerful numbers satisfies $n \bmod 36 \in \{7, 27, 35\}$. Proposition 3 of [59] refines this: modulo ℓ^2 the proportion of admissible residues is $1/4$ for $\ell = 2$, $1/3$ for $\ell = 3$ and $(\ell^2 - 3\ell + 3)/\ell^2$ for $\ell \geq 5$, and the density of admissible residues tends to 0 like $(\log L)^{-3}$. Every factor is positive, so for every finite modulus some residue class survives, and no finite sieve of this type decides the problem. Over the primes up to 10^5 the product of these proportions is about $2.67 \cdot 10^{-4}$. Congruences remain useful as search filters: by Corollary I.4.8(i), a prime of rank d satisfies $p \equiv \pm 1 \pmod{4d}$, so a search for such primes needs only 2 of the $2d$ odd residue classes modulo $4d$.

In words. Conditions modulo a fixed number can never rule out all triples: some remainders always survive. They thin out the candidates more and more, but never to zero. They are still good filters for computer searches.

Remark I.8.2 (Yokoi's representation). [computed] Mollin and Walsh [45, p. 112] suggest that Yokoi's representation of the fundamental unit [75, Proposition 1] may help. In our notation it gives, for a prime $p \equiv 3 \pmod{4}$ dividing both m and U_1 , that p^2 divides $T_1 - 1$ or $T_1 + 1$. The identity $T_1^2 - 1 = mU_1^2$ gives more: p^3 divides $(T_1 - 1)(T_1 + 1)$, and since p is odd, p^3 divides one of the two factors. So the representation adds no new constraint. We checked this on all 123 such pairs (m, p) with T_1 even and $m \leq 20000$.

In words. An idea that Mollin and Walsh suggested gives nothing new here: the condition it yields already follows from the equation $T_1^2 - 1 = mU_1^2$.

Remark I.8.3 (small units). [computed] A search beyond Theorem I.3.1 might target kernels with a small fundamental unit, where many indices fit below a height bound. The kernels $m = t^2 - 2$ with t odd and $m = t^2 - 1$ with $4 \mid t$ have the units $(t^2 - 1) + t\sqrt{m}$ and $t + \sqrt{m}$. There $U_1 = t$ with t odd, resp. $U_1 = 1$, is prime to m , so $m' = m$, and by Theorem I.2.1 the

first admissible index $k = m$ is already far out. Among the 251 such squarefree kernels $m \equiv 7 \pmod{8}$ with $t \leq 399$, only 17 have a first candidate $T_m(m)$ with fewer than 2000 digits, and all of them have $m \leq 1023$, inside the range of Theorem I.3.1. For these two kinds of kernel, small units therefore bring no new kernels into reach.

In words. A small unit sounds helpful, but it makes m' as large as m , and then the first possible middle is enormous. For the two kinds of kernel we checked, nothing new appears.

The route through the *abc* conjecture is discussed in Remark I.4.12: the uniform explicit exponents are too weak, and the full explicit form gives only a conditional bound.

Remark I.8.4 (the polynomial analog). [literature] Over the polynomials in one variable over a field of characteristic 0, not even two consecutive nonconstant powerful polynomials exist. If f and $f + 1$ are both powerful of degree $n \geq 1$, the theorem of Mason and Stothers [68, 40], applied to $(f + 1) - f = 1$, gives $n \leq \deg \text{rad}(f(f + 1)) - 1 \leq n/2 + n/2 - 1$, a contradiction; see Snyder [63] for a short proof of the theorem. In the integers there are infinitely many consecutive powerful pairs, such as $(8, 9)$ and $(288, 289)$ [28]. The proof uses the derivative, which has no analog for integers. In characteristic p the statement fails as well: x^p and $x^p + 1 = (x + 1)^p$ are consecutive p -th powers, because the derivative of a p -th power vanishes.

In words. For polynomials the problem is easy, even for pairs, because one can take derivatives. For whole numbers there is no derivative, and pairs of consecutive powerful numbers do exist. So the polynomial proof cannot be copied.

9 Open questions

Question I.9.1 (finiteness of the kernels of wing (ii)). [open] Are there only finitely many squarefree $m \equiv 7 \pmod{8}$ with $m \mid U_1(m)$ and $T_1(m)$ even?

These are the kernels of wing (ii) in Corollary I.2.4, the weakest point for an unconditional bound (Section 7). Walsh [72, p. 97] asks the corresponding question for all squarefree m with $m \mid U_1(m)$.

Question I.9.2 (the primitive part of $T_d(m)$). [open] Let $m \equiv 7 \pmod{8}$ be squarefree with $T_1(m)$ even, let $d \geq 5$ be odd, and let M_d be the part of $T_d(m)$ supported on the primes of rank exactly d . Can M_d be powerful?

The case of a prime d is, in our view, the hardest, since Corollary I.4.8 then gives a single condition. Section 6 examines the question on a finite set of pairs (m, d) . For the Fibonacci numbers, Fathi [25, Theorems 1.2–1.6] shows for several infinite families of indices n , among them every prime power $n > 2$, that F_n has a prime factor that first appears at n and occurs to an odd power. For these n the primitive part is not a square; this is weaker than not being powerful. Granville [30, Theorem 3] proves, for recurrences $x_{n+2} = bx_{n+1} + cx_n$ with $c \equiv 2 \pmod{4}$ and further conditions, that the terms have a primitive prime factor to an odd power. Our recurrence $T_{n+2} = 2T_1T_{n+1} - T_n$ has $c = -1$, and the question here asks for exponent one, not for an odd exponent.

Question I.9.3 (a middle that is a power of 2). [open] Can $2^a - 1$ and $2^a + 1$ both be powerful for some $a \geq 2$?

This is the case of a middle that is a power of 2, the case that Granville’s construction produces (Section 4). Proposition I.6.7 answers it for all exponents up to a bound. She [61, p. 7] conjectures that $x^n - 1$ is never powerful for $x > 1$ and $n > 2$, which would answer the question.

Question I.9.4 (Reinhart’s condition (C)). [open] Does Reinhart’s condition (C) [53, Definition 1.4] hold for some squarefree d , that is, is there an index k with $T_k(d)$ powerful and $d \mid U_k(d)$? For $d \mid U_1(d)$ the second condition holds for every k .

Reinhart asks this in particular for $d = 4\,099\,215$ and $d = 39\,028\,039\,587\,479$, and even for $d = 7$ [53, p. 6]. For odd k , Theorems I.5.1 and I.7.1(vii) show how large such an index would have to be.

An expectation of the author, not a claim. If a triple exists, we expect it to belong to a special case that is singled out by an additional equation, such as a middle that is a proper power, rather than to a lattice point without such structure. The reason is the model of Proposition I.6.1: at a point without additional structure every condition of Corollary I.4.8 is one more rare event, so a triple there would need many independent coincidences, while an algebraic identity can satisfy several conditions at once, as the Pell equations do for pairs. The special shapes treated so far, the cube middles of [17, 61, 59], the square-type middles of Theorem I.5.4 and the powers of 2 of Proposition I.6.7, are excluded or restricted. The expectation says where a triple would most likely be found, not whether one exists.

In words. These are the questions this part leaves open. The first asks whether the dangerous kernels of wing (ii) are finite in number. The second asks whether a building block of the middle can be powerful. The third asks whether the middle can be a power of 2, which Proposition I.6.7 rules out for small exponents. The fourth is Reinhart’s question for single kernels, including the three of Table 3. Whether the structure forces an odd witness in an outer number is settled by Remark I.4.11: it does not, beyond the primes of $T_1 \pm 1$.

The type of every statement

Every statement of this part carries one of six labels, printed after its number. The table lists them; it is generated from the text by the build and checked against a second, independent reading of the source.

statement	title	type
Lemma I.1.1	Mollin and Walsh [45, Theorem, p. 110]	literature
Lemma I.1.2	Walker [71, Theorems 2.2(3) and 2.5]; Mollin and Walsh [45, p. 111]	literature
Lemma I.1.3	Sentance [60, Theorem and Converse, p. 272]	literature
Lemma I.1.4	Sentance [60, p. 272]; Mollin and Walsh [44, Lemma 1.1]	literature
Theorem I.2.1	height inequality	proved
Remark I.2.2	the constant in the height bound	proved
Corollary I.2.3	kernel bound	proved
Corollary I.2.4	two wings	proved
Remark I.2.5	Reinhart’s list	literature
Theorem I.3.1	reduction	computed
Remark I.3.2	weaker bounds with less work	proved

statement	title	type
Lemma I.4.1	Bennett and Walsh [8, proof of Lemma 3.3, p. 3484]; Lehmer [35]	literature
Lemma I.4.2	Bennett and Walsh [8, Theorem 1.2]	literature
Lemma I.4.3	valuation identity	proved
Proposition I.4.4	exponent-3 condition	proved
Theorem I.4.5	the b -box	computed
Theorem I.4.6	reduction to a witness statement	proved
Lemma I.4.7	the T_1 -part of a lattice point	proved
Corollary I.4.8	the divisor count of the Wieferich conditions	proved
Theorem I.4.9	Granville [29, Theorem, p. 216]	literature
Remark I.4.10	what the witness condition does not follow from	proved
Remark I.4.11	where a witness in an outer number can sit	proved
Remark I.4.12	two elementary constraints, and the size of the gap	proved
Remark I.4.13	the cyclotomic axis	proved
Theorem I.5.1	the strip	computed
Remark I.5.2	the bound of Chim, Shorey and Sinha	literature
Theorem I.5.3	smooth kernels	computed
Theorem I.5.4	square-type middles	computed
Remark I.5.5	the share of the candidates covered	computed
Proposition I.6.1	exponents at the rank	measured
Proposition I.6.2	no triple below $1.83 \cdot 10^{18}$, and the pairs at distance two with middle divisible by four	computed
Proposition I.6.3	ladders for the known kernels dividing U_1	computed
Proposition I.6.4	the smallest witnesses	computed
Remark I.6.5	the condition modulo 36 and the witnesses	computed
Proposition I.6.6	how many conditions there are, and how often they are met	computed
Proposition I.6.7	middles that are powers of two	computed
Proposition I.6.8	the primitive part over an explicit finite set	computed
Remark I.6.9	where this stands in the literature	literature
Proposition I.6.10	Wieferich primes of the unit and the primitive parts	proved
Remark I.6.11	the Wieferich events of the units	computed
Remark I.6.12	Lucas–Wieferich primes	computed
Remark I.6.13	a second primitive prime factor	computed
Theorem I.7.1	profile	computed
Remark I.8.1	congruences	literature
Remark I.8.2	Yokoi’s representation	computed
Remark I.8.3	small units	computed
Remark I.8.4	the polynomial analog	literature
Question I.9.1	finiteness of the kernels of wing (ii)	open
Question I.9.2	the primitive part of $T_d(m)$	open
Question I.9.3	a middle that is a power of 2	open
Question I.9.4	Reinhart’s condition (C)	open

The types: *proved*, proved in this part; the proof may use published results (15); *literature*, due to the cited source, sometimes with a proof given here for completeness (12); *computed*, the result of an exact computation, or a proof that uses one of our computations (19); *measured*, a measurement, with a model that is not proved (1); *open*, an open question (4).

Data and code availability

Every number of this part that we computed is generated from a result file of a data deposit or from a cited constant; no computed number is typed by hand. The data deposit is archived with the collected volume, DOI 10.5281/zenodo.23127547.

What the deposit contains. The deposit has two parts. The core holds what the text cites: (i) the result files, in JSON and plain text, behind every computed number, together with the raw output of the scripts that wrote them; (ii) the scripts, written in Python, each stating before it runs what it expects to find and the positive and negative controls that it runs; (iii) a manifest with the SHA-256 hash and the size of every file, which the build checks before it reads a result file; (iv) the decimal expansions of the building blocks of Part I that the searches left open (Proposition I.6.8), including those that the algebraic split of Remark I.6.13 settles; (v) the certificates and the second routes for the computations that settle a building block; and (vi) the file `OPEN_PROBLEMS.md`, which lists the open problems of the volume. The archive holds the remaining scripts and results of the investigation, including negative results that the text does not print.

Where the files are. The collected volume and its three parts are published as four linked records on Zenodo, and the same files are in one repository on GitHub. The record of the collected volume (DOI 10.5281/zenodo.23127547) holds the volume and the whole deposit, the core and the archive described above. The record of each part holds that part and the result files that its text cites, and links to the record of the volume. The repository (github.com/berkay-yueksel-sayim/erdos-364-365-powerful-numbers-pell) holds the four PDFs and the deposit, and its README lists the four records. The records of Parts I, II and III have the DOIs 10.5281/zenodo.23127703, 10.5281/zenodo.23127705 and 10.5281/zenodo.23127707.

How the numbers get into the text. The source of the text refers to each computed number by a key. A build tool reads the result file named for the key, verifies it against the manifest, computes the number in its printed form, and writes it into the text; it stops if a key has no source or if a file differs from the manifest. The same tool generates the table of the types of the statements and the list of open problems from the source of the text, and it is part of the core.

Software. The scripts use Python 3.12 with gmpy2 2.3, SymPy 1.14 and NumPy 2.4. The elliptic curve method and its relatives are run with GMP-ECM 7.0.6 [76], used as an installed program inside a container image built from the Debian package and not part of the deposit; the recipe of the image is in the core. The elliptic curve certificates for primality were produced in the same way with PARI/GP 2.17.2 [70] and are part of the deposit; a separate program of ours checks each of them, so the result does not depend on that program. Further work is planned to support these certificates with a second prover written independently by us. The text is set with pdf $\text{\TeX}$.

Licenses. The text and the data are licensed under the Creative Commons Attribution 4.0 International License (CC BY 4.0), and the code under the Apache License, Version 2.0. Programs of others that the scripts use as tools, such as GMP-ECM, are not part of the deposit and keep their own licenses.

Use of AI tools

Generative AI tools were used throughout this work: large language models of the Claude family by Anthropic, mainly Claude Opus 5 and 5.5, and also Claude Fable 5.1 and Claude Sonnet 5.5 and 5, between August and October 2026, through the Claude Code environment. They were used to draft the text and the mathematical arguments, to write the computer programs, to run and check the computations, and to search for and read the literature. The author works independently, is not trained as a number theorist, and learned the subject actively during the work; the tools made it possible to carry out and check computations on this scale. The author led the work in stages: each stage was set by a prompt that stated what it should achieve, within a stage the tools followed a plan agreed with the author, and the author considered the results before giving the next prompt, revising the plan when the results required it. The author contributed ideas where they helped and framed how a problem should be attacked, decided which statements to keep, and is responsible for the content. The programs of the project, including its own checking and build tools, were developed with the AI tools, and the author started the long computations, such as the factor searches, on the author's own machine.

Several parts of the checking do not rest on these tools. The elliptic curve primality certificates were produced by PARI/GP 2.17.2 and are verified by a separate program, and the factor searches use GMP-ECM 7.0.6. Every computed number is generated by a program from deposited data or from a cited constant, and the programs carry positive and negative controls. Proofs, numbers and citations were also checked in independent passes by further instances of the AI tools that were given no access to the drafting context. These tools are not authors.

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